
Non-parametric E(3)-equivariant local descriptors for electronic Hamiltonian prediction

Research, mathematical, implementation, and experimental specification

Target platform: MANDALA

Primary benchmark: HamGNN SiO₂

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Target Hamiltonian:	spinless, non-SOC, self-consistent H^{SCF}
Model class:	deterministic local descriptors + pairwise equivariant mapper
Learned message passing:	prohibited
Primary model output:	complete Hamiltonian block irrep vector
Project storage budget:	2 TB total

Executive specification

This document defines a controlled research program for replacing learned message-passing neighborhoods with precomputed, non-parametric, E(3)-equivariant atomic descriptors of known irreducible-representation content. For every Hamiltonian pair (i, j) inside a dataset-specific Hamiltonian cutoff R_H , a learned pair mapper receives only two precomputed endpoint descriptors and the pair displacement, and predicts the *complete* irreducible-representation vector of the corresponding self-consistent Hamiltonian block. A frozen dataset-specific range-magnitude envelope is applied before the primary loss.

The scientific objective is not only to maximize accuracy. It is to determine which descriptor classes are geometrically complete or stably near-complete at practical truncation, which pairwise equivariant mapping families use that information efficiently, and at what descriptor radius R_D the local input becomes sufficient for predicting H_{ij} . The program therefore separates three questions that are often conflated: descriptor injectivity, target sufficiency, and learnability.

Non-negotiable project invariants

- The target is the self-consistent spinless non-SOC Hamiltonian H^{SCF} in the canonical MANDALA/OpenMX AO convention.
- Every pair mapper returns one complete Hamiltonian-block irrep vector. Shell labels may organize internal multiplicities or diagnostics but do not define separate prediction heads.
- There is no learned graph message passing, iterative node update, or learned neighbor aggregation. Each pair prediction may access only precomputed endpoint descriptors, the pair displacement and fixed metadata.
- The primary training target is the irrep vector. The headline evaluation metric is MAE over reconstructed physical Hamiltonian matrix elements.
- R_H is established from each dataset using training data and an explicit omitted-tail criterion. R_D is independently swept and may exceed R_H .
- One frozen range-magnitude construction is used per dataset for every competing method. Its fit uses training data only.
- Descriptor bases, normalizers, envelopes, splits, and all other data-derived transforms are frozen from the training partition before validation/test evaluation.
- The first implementation stage is a native reimplement and re-evaluation of an ACEhamiltonians-style non-message-passing baseline, with the upstream package used only as a numerical reference.
- GNN/message-passing methods are not reimplemented; their published numbers are contextual references only.
- The total persistent storage footprint of the project may not exceed 2 TB.

Primary benchmark and references

The main public benchmark is the HamGNN SiO₂ dataset: 221 Materials Project SiO₂ structures, each perturbed three times by random atomic displacements of 0.1 Å, for 663 structures with cells up to 80 atoms. HamGNN reports a random 80/10/10 split and a test Hamiltonian-element MAE of 2.29 meV. It also reports 1.67 meV on a substantially larger 180-atom SiO₂ transfer structure. The first project phase will reproduce the same random-split *protocol*; the exact released split will be used if recoverable. A later robustness phase will introduce a parent-polymorph-grouped split.

The primary reproduced non-GNN baseline is an equivariant ACE Hamiltonian model. A second non-GNN baseline, scheduled after the common infrastructure is stable, is an N-center equivariant representation with symmetry-adapted linear/kernel regression. Monolayer MoS₂ from the DeepH-E3/SLEM benchmark family is the secondary public chemistry for final confirmation; published message-passing results are used only as reference numbers.

Program structure

The execution is divided into eight gated stages. Each stage has hypotheses, deliverables, acceptance criteria, expected outcomes, mandatory comparisons, and a standardized report. Critical symmetry, reference-agreement, data-integrity, or leakage failures block progression.

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1 Scope, terminology, and research questions

1.1 Model boundary

For a periodic structure $\mathcal{C}$ with atoms $i = 1, \dots, N$, Cartesian positions $\mathbf{R}_i$, species Z_i , and lattice translations $\mathbf{L}$, define a directed pair displacement

$$\mathbf{r}_{ij}^{\mathbf{L}} = \mathbf{R}_j - \mathbf{R}_i + \mathbf{L}, \quad r_{ij}^{\mathbf{L}} = \|\mathbf{r}_{ij}^{\mathbf{L}}\|. \quad (1)$$

A Hamiltonian pair is included when $r_{ij}^{\mathbf{L}} \leq R_H$, with special treatment for the onsite block $i = j, \mathbf{L} = \mathbf{0}$. For every atom, a deterministic descriptor is precomputed once,

$$D_i = D(\mathcal{N}_i; R_D), \quad (2)$$

where $\mathcal{N}_i$ contains the species-labelled neighborhood inside descriptor radius R_D . The learned offsite map is

$$F_\theta : (D_i, \mathbf{r}_{ij}^{\mathbf{L}}, D_j, Z_i, Z_j) \mapsto \tilde{\mathbf{h}}_{ij}^{\mathbf{L}}, \quad (3)$$

followed by a fixed range-magnitude transform and exact block reconstruction,

$$\hat{\mathbf{h}}_{ij}^{\mathbf{L}} = G_{ij}(r_{ij})\tilde{\mathbf{h}}_{ij}^{\mathbf{L}}, \quad \hat{H}_{ij}^{\mathbf{L}} = Q_{ij}^\top \hat{\mathbf{h}}_{ij}^{\mathbf{L}}. \quad (4)$$

Here $\mathbf{h}_{ij}$ denotes the *entire* irrep vector of the AO block, not an individual shell-pair target.

The onsite map is separate because $\mathbf{r}_{ii} = 0$ provides no bond axis:

$$F_\theta^{\text{on}} : (D_i, Z_i) \mapsto \tilde{\mathbf{h}}_{ii}^{\mathbf{0}}. \quad (5)$$

Shifted-self periodic blocks $i = j, \mathbf{L} \neq 0$ are ordinary nonzero-displacement pair inputs.

1.2 What is forbidden and what is allowed

A deterministic descriptor may sum or contract functions of direct neighbors because that is part of precomputation. A pair mapper may form arbitrary symmetry-valid functions of its two cached endpoint descriptors and pair geometry. It may not update an atom representation using learned information from a graph and then reuse that updated representation for other pairs. In particular, graph scatter operations, learned neighborhood aggregation, iterative edge/node blocks, and recurrent propagation are outside scope.

This boundary still permits rich local many-body information. A descriptor can contain high-order density correlations, moments, tensor products, or frequency components, provided they are deterministic once basis hyperparameters are fixed. Likewise a pair mapper may contain deep equivariant tensor-product or local-frame networks: depth in a pairwise function is allowed; learned propagation across atoms is not.

1.3 Core research questions

1. Which finite, non-parametric, typed-O(3) descriptor best balances geometric information, stable invertibility, storage, and precomputation cost?
2. Under what explicit assumptions can at least one finite descriptor be proved injective, and how close are practical truncations of the other descriptors to that standard?
3. Does geometric descriptor completeness correlate with Hamiltonian accuracy, or does prediction saturate earlier because only a lower-dimensional target-sufficient quotient is required?
4. At what R_D is $(D_i, \mathbf{r}_{ij}, D_j)$ sufficient for H_{ij} at the chosen DFT protocol, and what irreducible error remains from physics outside the cutoff?
5. Which equivariant pair mapper achieves the best accuracy/cost frontier: dense global Clebsch-Gordan products, fixed polynomial covariants, invariant reduced-coefficient networks, factorized/sparse products, Gaunt products, or bond-frame O(2) processing?
6. Can local O(2) processing reproduce the accuracy of full O(3) tensor products while reducing high- ℓ cost, including pseudo-irrep channels required by general AO Hamiltonians?
7. Does deterministic high-order feature lifting improve sample efficiency enough to justify its storage footprint, or is it better to keep a compact complete descriptor and learn nonlinear pair couplings on demand?

8. How do these non-message-passing models compare with reproduced non-GNN baselines and published GNN accuracies on the same chemistry?

2 Representation theory and Hamiltonian targets

2.1 From E(3) to typed O(3) irreps

Translation invariance is obtained by using relative coordinates. The remaining spatial action is O(3), including proper rotations and inversion/reflections. An O(3) irrep is labelled by angular momentum $\ell \in \mathbb{N}_0$ and parity $p \in \{+1, -1\}$ and has dimension $2\ell + 1$:

$$x^{\ell p} \mapsto D^{\ell p}(Q)x^{\ell p}, \quad Q \in \text{O}(3). \quad (6)$$

We use e3nn-style labels $0e, 0o, 1e, 1o, \dots$, where “e” is even and “o” is odd under inversion. A complete feature vector is a direct sum

$$x \in \bigoplus_{\ell, p} m_{\ell p} \mathcal{V}_{\ell p}, \quad (7)$$

where $m_{\ell p}$ is a multiplicity index. Learned channel mixing may act freely in multiplicity space but must not arbitrarily mix the $m = -\ell, \dots, +\ell$ magnetic components of one irrep copy.

Useful examples are

Type	Physical interpretation	inversion
0e	scalar	unchanged
0o	pseudoscalar	sign reversal
1o	polar vector	sign reversal
1e	axial vector / pseudovector	unchanged
2e, 2o	rank-2 irreducible tensors	parity-labelled

2.2 Spherical harmonics and bond geometry

A bond direction admits the real spherical-harmonic expansion

$$Y_{\ell m}(\hat{\mathbf{r}}), \quad \hat{\mathbf{r}} = \mathbf{r}/r, \quad (8)$$

which transforms as natural parity $\ell^{(-1)^\ell}$. Radial features $R_n(r)$ are 0e. Bond basis functions are therefore

$$b_{n\ell m}(\mathbf{r}) = R_n(r)Y_{\ell m}(\hat{\mathbf{r}}). \quad (9)$$

Spherical harmonics encode direction while radial functions encode invariant range. For $\mathbf{r} = 0$, all nonscalar directional components are undefined; the onsite model must not invent a direction.

2.3 Clebsch-Gordan products

For irreps (ℓ_1, p_1) and (ℓ_2, p_2) ,

$$\mathcal{V}_{\ell_1 p_1} \otimes \mathcal{V}_{\ell_2 p_2} = \bigoplus_{L=|\ell_1-\ell_2|}^{\ell_1+\ell_2} \mathcal{V}_{L, p_1 p_2}. \quad (10)$$

The projected product is

$$[x^{\ell_1 p_1} \otimes y^{\ell_2 p_2}]_{LM} = \sum_{m_1 m_2} C_{\ell_1 m_1, \ell_2 m_2}^{LM} x_{m_1} y_{m_2}. \quad (11)$$

Clebsch-Gordan coefficients are fixed. Learnable parameters may mix legal paths and multiplicities, but the allowed angular momenta and parity are analytic constraints.

2.4 Why arbitrary componentwise nonlinearities are invalid

For a nonscalar irrep x^ℓ , a componentwise function σ generally fails

$$\sigma(D^\ell(Q)x) \neq D^\ell(Q)\sigma(x). \quad (12)$$

Valid nonlinear constructions include invariant norms, 0e scalar gates, tensor products, spherical-domain nonlinearities with a correct transform, and invariant-conditioned linear modulation. For a pseudoscalar s^{0o} , a pointwise scalar activation must be odd, $f(-s) = -f(s)$, if it is to remain 0o. Only 0e channels may receive arbitrary additive biases.

2.5 Full Hamiltonian block irreps

Consider a real spinless AO basis with orbital angular momenta ℓ_a on atom i and ℓ_b on atom j . A shell-pair block transforms in

$$\mathcal{V}_{\ell_a, (-1)^{\ell_a}} \otimes \mathcal{V}_{\ell_b, (-1)^{\ell_b}} = \bigoplus_{L=|\ell_a-\ell_b|}^{\ell_a+\ell_b} \mathcal{V}_{L, (-1)^{\ell_a+\ell_b}}. \quad (13)$$

The parity of every L sector for that shell pair is fixed by $(-1)^{\ell_a+\ell_b}$, not by $(-1)^L$. Thus pseudo irreps arise naturally:

orbital pair	target irreps
s - s	0e
s - p	1o
p - p	0e $\oplus$ 1e $\oplus$ 2e
p - d	1o $\oplus$ 2o $\oplus$ 3o
d - d	0e $\oplus$ 1e $\oplus$ 2e $\oplus$ 3e $\oplus$ 4e

Consequently a natural-parity descriptor containing only 0e, 1o, 2e, ... is not by itself a complete target basis. Nonlinear tensor products or precomputed covariants must be able to create, for example,

$$1o \otimes 1o = 0e \oplus 1e \oplus 2e. \quad (14)$$

2.6 MANDALA block-to-irrep map and the full-block interface

Let H_{ij} denote the complete AO block after the canonical AO ordering is fixed. MANDALA supplies a fixed linear map Q_{ij} that converts the block to a direct sum of target irreps:

$$\mathbf{h}_{ij} = Q_{ij} H_{ij}, \quad H_{ij} = Q_{ij}^\top \mathbf{h}_{ij} \quad (15)$$

when Q_{ij} is orthogonal under the chosen vectorization; otherwise the stored inverse is used explicitly. The project will verify this property numerically rather than assume it.

Hard API invariant: one complete block

For a chemical pair and AO basis, define a fixed ordered target schema

$$\mathcal{I}_{ij} = \bigoplus_{q=1}^{M_{ij}} \mathcal{V}_{L_q p_q}.$$

The multiplicity index q implicitly contains every radial/shell-pair occurrence belonging to the block. A mapper receives one pair input and returns one vector in $\mathcal{I}_{ij}$. There are no independently trained shell-pair heads. Internal code may group equal (L, p) copies for efficiency, and diagnostics may report shell-resolved errors after reconstruction, but these are implementation views of a single block prediction.

2.7 Hermiticity and pair reversal

For a real spinless periodic Hamiltonian,

$$H_{ji}^{-\mathbf{L}} = (H_{ij}^{\mathbf{L}})^\top. \quad (16)$$

In the irrep basis this induces a fixed linear map

$$\mathbf{h}_{ji}^{-\mathbf{L}} = S_{ij} \mathbf{h}_{ij}^{\mathbf{L}}, \quad (17)$$

including angular exchange phases and radial/shell multiplicity permutation. Under standard CG conventions an angular factor $(-1)^{\ell_a+\ell_b-L}$ appears, but the implementation must derive S_{ij} from the exact MANDALA basis conventions and test it, not hard-code a generic sign formula.

Two allowed enforcement strategies are:

1. **Canonical pair prediction:** predict one canonical directed representative and generate the reverse block analytically with S_{ij} .
2. **Hermitian projection:** evaluate both directions and project

$$\mathbf{h}_{ij}^{\text{proj}} = \frac{1}{2} (\mathbf{h}_{ij} + S_{ji} \mathbf{h}_{ji}). \quad (18)$$

Canonical prediction is the production default; projection is retained as a diagnostic and ablation.

3 Completeness, locality, and target sufficiency

3.1 Species-resolved neighborhood measure

Represent the neighborhood of atom i as species-resolved finite point measures

$$\mu_i^{(s)} = \sum_{\substack{a \neq i \\ Z_a = s \\ r_{ia} < R_D}} w(r_{ia}) \delta_{\mathbf{r}_{ia}}, \quad \mathbf{r}_{ia} = \mathbf{R}_a - \mathbf{R}_i + \mathbf{L}_{ia}. \quad (19)$$

The periodic image convention must be deterministic and stored. Species remain separate channels. A scalar numerical encoding such as replacing one-hot species by Z_a is not acceptable for completeness claims because different species contributions may cancel in moments.

3.2 Equivariance and injectivity definitions

A descriptor is permutation-invariant over equal-species neighbors and $O(3)$ -equivariant if

$$D(Q\mathcal{N}_i) = \rho_D(Q)D(\mathcal{N}_i), \quad Q \in O(3). \quad (20)$$

The strongest useful local notion is frame-aware injectivity:

$$D(\mathcal{N}) = D(\mathcal{N}') \implies \mathcal{N} = \mathcal{N}' \quad (21)$$

up to permutations of equal-species atoms, with both neighborhoods in the same global frame. Orbit completeness is weaker:

$$D(\mathcal{N}') = \rho_D(Q)D(\mathcal{N}) \iff \mathcal{N}' = Q\mathcal{N}. \quad (22)$$

3.3 Invariant endpoints plus a bond vector are not pair-complete

Suppose I_i and I_j are complete *invariant* endpoint descriptors and the pair input is $(I_i, \mathbf{r}_{ij}, I_j)$. Rotate the non-axisymmetric neighborhood around i by an arbitrary angle about the axis $\mathbf{r}_{ij}$ while leaving j unchanged. I_i , I_j , and $\mathbf{r}_{ij}$ are unchanged, but the relative torsional orientation of the two endpoint environments changes. A Hamiltonian block may respond to this torsion. Therefore complete invariant endpoints plus a bond vector are not generally complete for the pair geometry.

The endpoint representation must retain covariant orientation, or the pair representation must explicitly encode the missing relative frame information.

3.4 Pair-completeness proposition

Proposition 1. Assume a frame-aware descriptor D admits a decoder $\mathcal{R}$ such that $\mathcal{R}(D_i) = \mu_i$ for every neighborhood in a specified domain. Then the tuple $(D_i, \mathbf{r}_{ij}, D_j)$ determines the union $\mathcal{N}_i \cup \mathcal{N}_j$ in a common frame, up to equal-species permutations.

Reason. Decode both endpoint neighborhoods. Coordinates in j 's centered frame are translated into i 's frame by

$$\mathbf{r}_{ia} = \mathbf{r}_{ij} + \mathbf{r}_{ja}. \quad (23)$$

Thus the relative orientation of the two decoded point sets is fixed. Overlapping atoms can be identified by species and position. No independent rotation of either neighborhood remains. $\square$

If the target obeys a strict locality assumption

$$H_{ij} = \mathcal{H}(\mathcal{N}_i \cup \mathcal{N}_j, \mathbf{r}_{ij}; \xi) \quad (24)$$

for fixed electronic-state settings ξ , a deterministic pairwise map exists in principle. This is a statement about existence, not about practical smoothness or learnability.

3.5 Why finite completeness needs a restricted domain

A generic center-fixed neighborhood with N neighbors has $3N$ continuous degrees of freedom. After quotienting by global proper rotations it has generically $3N - 3$. Therefore a continuous fixed-frame descriptor of dimension $d < 3N$ cannot generically be injective on an open set; an invariant descriptor needs at least approximately $3N - 3$ continuous coordinates. A fixed finite descriptor cannot be complete for unbounded neighbor count.

Any finite-dimensional theorem must therefore state an explicit domain, for example

$$\mathfrak{C}(N_{\max}, R_D, \delta_{\min}, \delta_R, \mathcal{S}), \quad (25)$$

with at most $N_{\max}$ neighbors, finite species set $\mathcal{S}$, pair separation at least $\delta_{\min} > 0$, and all neighbor radii bounded by $R_D - \delta_R$. The cutoff margin matters because a window with $w(R_D) = 0$ makes an atom exactly at the boundary invisible and creates severe conditioning near the boundary.

3.6 Geometric completeness versus Hamiltonian sufficiency

Define the complete pair input

$$z_{ij}(\mathcal{C}) = (D_i, D_j, \mathbf{r}_{ij}, Z_i, Z_j, \xi), \quad (26)$$

and target full-block irrep vector $\mathbf{h}_{ij}(\mathcal{C})$. Target sufficiency means there exists a deterministic map F satisfying

$$\mathbf{h}_{ij} = F(z_{ij}), \quad (27)$$

or equivalently

$$z_{ij}(\mathcal{C}) = z_{ij}(\mathcal{C}') \implies \mathbf{h}_{ij}(\mathcal{C}) = \mathbf{h}_{ij}(\mathcal{C}'). \quad (28)$$

Geometric completeness is neither necessary nor sufficient for this statement. An incomplete descriptor can still preserve all target-relevant degrees of freedom; a geometrically complete finite-radius descriptor can fail when the self-consistent Hamiltonian depends measurably on matter or fields outside the radius.

The project fixes the electronic-state protocol ξ : same functional, pseudopotentials, AO basis, charge/spin settings, smearing/occupations, and boundary conventions. Any variable allowed to change must be added to z_{ij} with its proper symmetry type.

3.7 Differential necessity condition for target sufficiency

Let $\mathbf{q}$ collect structural degrees of freedom in a differentiable neighborhood of a configuration, and define

$$J_z = \frac{\partial z_{ij}}{\partial \mathbf{q}}, \quad J_h = \frac{\partial \mathbf{h}_{ij}}{\partial \mathbf{q}}. \quad (29)$$

If $\mathbf{h} = F(z)$ locally, then

$$J_h = J_F J_z, \quad (30)$$

so necessarily

$$\ker J_z \subseteq \ker J_h. \quad (31)$$

With $P_{\ker J_z} = I - J_z^+ J_z$, define the missed-target sensitivity

$$\eta_{\text{miss}} = \frac{\|J_h P_{\ker J_z}\|_F}{\max(\|J_h\|_F, \epsilon)}. \quad (32)$$

A statistically significant nonzero value proves a local target-relevant direction is invisible to the descriptor input. This is a model-free failure certificate. J_z can be obtained analytically/autodiff; J_h requires finite-difference electronic-structure calculations or DFPT and must be compared against an SCF/gauge noise floor.

4 Descriptor families

Four primary descriptor classes will be implemented. They are deliberately related: all can be understood as coordinate systems or deterministic nonlinear lifts of the same species-resolved point measure. The comparison is therefore about finite truncation, stable inversion, computational conditioning, storage, and how readily a mapper can use the information.

4.1 D1: raw species-resolved neighbor-density coefficients

Define a species-resolved local density

$$\rho_i^{(s)}(\mathbf{x}) = \sum_{a:Z_a=s} w(r_{ia}) g_\sigma(\mathbf{x} - \mathbf{r}_{ia}), \quad (33)$$

where g_σ may be a delta distribution or a fixed smoothing kernel. Expand in an orthogonal radial-spherical basis:

$$c_{sn\ell m}^{(i)} = \int R_{n\ell}(r) Y_{\ell m}^*(\hat{\mathbf{x}}) \rho_i^{(s)}(\mathbf{x}) d\mathbf{x}. \quad (34)$$

For delta atoms,

$$c_{sn\ell m}^{(i)} = \sum_{a:Z_a=s} w(r_{ia}) R_{n\ell}(r_{ia}) Y_{\ell m}^*(\hat{\mathbf{r}}_{ia}). \quad (35)$$

Each fixed (s, n, ℓ) channel transforms as $\ell^{(-1)^\ell}$, so D1 is a direct sum of $0e, 1o, 2e, 3o, \dots$

4.1.1 Completeness argument

If the radial-angular basis is complete in the ball, the coefficient sequence reconstructs $\rho_i^{(s)}$. If g_σ is a delta, its support directly gives all neighbor positions and species. If a Gaussian kernel is used, convolution remains injective in the complete representation because its Fourier transform is nowhere zero; exact deconvolution exists in principle. A strictly positive known cutoff weight can be divided out after support recovery. Therefore the infinite coefficient sequence is an injective frame-aware encoding of the colored neighborhood.

This is a complete-basis statement. A finite truncation ($n \leq n_{\max}, \ell \leq \ell_{\max}$) requires independent reconstruction and collision certification.

4.1.2 Radial basis candidates

The first implementation should support at least two fixed bases behind one interface:

- orthogonal spherical-Bessel/Jacobi basis on $[0, R_D]$;
- 3D Zernike-compatible orthogonal radial functions.

Raw high-degree monomials are not the production default because of dynamic-range and conditioning problems. Basis functions and normalization constants are fixed buffers, not learned parameters.

4.1.3 Pair puncturing

D1 is additive. For an offsite pair the explicitly supplied partner may be removed analytically:

$$D_{i \setminus j} = D_i - \phi(Z_j, \mathbf{r}_{ij}), \quad D_{j \setminus i} = D_j - \phi(Z_i, -\mathbf{r}_{ij}). \quad (36)$$

Three inputs will be compared: full descriptors, punctured descriptors, and the concatenation of both. Puncturing must respect periodic image identity, not merely species and nearest distance.

4.2 D2: irreducible moments and Zernike/STF tensors

Define species-resolved Cartesian moments

$$M_{s,q,\nu}^{(i)} = \sum_{a:Z_a=s} w(r_{ia}) f_q(r_{ia}) \mathbf{r}_{ia}^{\otimes \nu}. \quad (37)$$

A symmetric rank- ν Cartesian tensor decomposes into symmetric-traceless irreps

$$\ell = \nu, \nu - 2, \dots, \quad p = (-1)^\nu. \quad (38)$$

Equivalently store normalized regular-solid-harmonic or 3D Zernike moments

$$Z_{sn\ell m}^{(i)} = \sum_{a:Z_a=s} w(r_{ia}) R_{n\ell}^Z(r_{ia}) Y_{\ell m}^*(\hat{\mathbf{r}}_{ia}). \quad (39)$$

The STF/Zernike transformation is a fixed invertible change of basis from the associated Cartesian moment collection, so no information is lost when all required components are retained.

4.2.1 Finite-support completeness route

For compactly supported finite point measures, sufficiently rich polynomial moments determine the measure. Under bounded support size and non-collision/separation assumptions, multivariate Prony-type recovery provides a path to a finite uniqueness theorem. D2 is the primary theorem-oriented descriptor: the project will attempt to state and prove an injectivity result on

$$\mathfrak{C}(N_{\max}, R_D, \delta_{\min}, \delta_R, \mathcal{S}) \quad (40)$$

for a concrete finite moment/Zernike set. If a fully global finite theorem proves too restrictive, the deliverable must precisely identify whether the result is generic, local, or stability-qualified; empirical certification cannot be described as a theorem.

4.2.2 Implementation preference

Internally normalize radii as $\tilde{r} = r/R_D \in [0, 1]$ and use an orthogonal basis. Analytic recurrence and analytic descriptor Jacobians should be implemented. Raw Cartesian moments may be exposed only as a validation view for comparison against the irreducible representation.

4.3 D3: Fourier-Bessel characteristic-function coefficients

For each species define the characteristic function of the local point measure,

$$\hat{\mu}_i^{(s)}(\mathbf{k}) = \sum_{a:Z_a=s} w(r_{ia}) e^{-i\mathbf{k} \cdot \mathbf{r}_{ia}}. \quad (41)$$

Using the plane-wave expansion yields spherical Fourier coefficients

$$F_{sq\ell m}^{(i)} = \sum_{a:Z_a=s} w(r_{ia}) j_\ell(qr_{ia}) Y_{\ell m}^*(\hat{\mathbf{r}}_{ia}), \quad (42)$$

where j_ℓ is a spherical Bessel function and q is a selected radial frequency. In a real basis, conventional i^ℓ factors are absorbed so stored values are real. Each (s, q, ℓ) channel again has natural parity $\ell^{(-1)^\ell}$.

4.3.1 Completeness argument and research opportunity

The complete characteristic function uniquely determines a finite measure, so continuous q and complete angular order are injective. Finite multidimensional Fourier samples can recover finite exponential sums under Prony-type conditions, but an arbitrary collection of spherical shells and finite $\ell_{\max}$ does not automatically inherit such a theorem. A substantive descriptor result would derive a finite shell/angular sampling scheme equivalent to a sufficient set of Cartesian Fourier moments for bounded $N_{\max}$ and separation $\delta_{\min}$.

D3 is attractive because Bessel functions are bounded, frequency resolution has a direct length-scale interpretation, and the descriptor connects naturally to periodic/reciprocal-space analysis. Its risk is aliasing or poor finite-shell conditioning if the sampling theorem is not respected.

4.4 D4: D1 plus deterministic ACE/NICE covariant lifts

Starting from D1 coefficients, construct selected higher-degree covariants using fixed Clebsch-Gordan contractions:

$$A_{\alpha,LP}^{(2)} = [c_{\alpha_1} \otimes c_{\alpha_2}]_{LP}, \quad (43)$$

$$A_{\alpha,LP}^{(3)} = [[c_{\alpha_1} \otimes c_{\alpha_2}]_K \otimes c_{\alpha_3}]_{LP}, \quad (44)$$

and analogously for higher body degree if later justified. The parity is the product of input parities. These lifts create target-relevant pseudo irreps, for example $[1o \otimes 1o]_{1e}$.

The production D4 representation retains D1 as a direct summand:

$$D_4 = D_1 \oplus A^{(2)} \oplus A^{(3)} \oplus \dots \quad (45)$$

Consequently D4 inherits any injectivity property of D1. The extra covariants are not claimed to add information; they expose nonlinear functions of existing information in a form a shallow pair mapper can exploit.

Deterministic path selection will use:

- angular triangle and parity rules;
- target-irrep reachability;
- maximum polynomial/body degree;
- fixed radial-degree budget;
- index/permutation symmetries;
- analytic norm/scale pruning that is independent of training labels.

PCA, supervised contraction, learned feature selection, or validation-driven path pruning are separate ablations and must not be included in the core claim of a non-parametric descriptor.

4.5 Exact ragged point-set oracle

Where feasible, cache

$$X_i = \{(Z_a, \mathbf{r}_{ia}, \mathbf{L}_{ia})\}_{a \in \mathcal{N}_i}. \quad (46)$$

This variable-length representation is trivially complete and serves as a reconstruction oracle, collision verifier, and optional upper-bound input. It is not one of the fixed-vector production descriptors. If used in a learned control model, processing must be a one-shot permutation-invariant/equivariant local set map without message passing.

4.6 Optional long-range field descriptor

If target-sufficiency tests reveal substantial outside-cutoff dependence that is smooth and multipolar, append a deterministic LODE-like local expansion of long-range potential fields,

$$D_i^{\text{aug}} = D_i^{\text{local}} \oplus D_i^{\text{LR}}. \quad (47)$$

This is a contingency, not part of the initial descriptor grid. It is activated only after Stage 6 demonstrates a measurable locality failure that cannot be removed by a practical increase in R_D .

4.7 Representations excluded as sole completeness carriers

SOAP power spectra, bispectra/SNAP, ACSFs, distance histograms, angle histograms, and finite low-body-order invariant contractions are not accepted as the only endpoint descriptor. Known counterexamples show finite low-order invariant correlations can map distinct environments to identical features; no downstream nonlinear mapper can recover information already discarded. Such invariants may be appended as 0e conditioning channels, but the main descriptor must retain frame-aware covariant information.

5 Training-independent descriptor certification

Descriptor quality will be evaluated before fitting Hamiltonians. No validation Hamiltonian error is allowed to serve as the sole criterion for selecting descriptor resolution.

5.1 Symmetry and PBC tests

For random $Q \in \text{O}(3)$, measure

$$\epsilon_{\text{eq}}(Q) = \frac{\|D(Q\mathcal{N}) - \rho_D(Q)D(\mathcal{N})\|}{\max(\|D(\mathcal{N})\|, \epsilon)}. \quad (48)$$

Test random proper rotations, inversion, at least two independent improper reflections, same-species permutations, cell translations, equivalent periodic-image remappings, and neighbor-list ordering. Run in float64 and intended production precision. A descriptor that passes only $\text{SO}(3)$ but fails parity is rejected for this project.

5.2 Analytic or constructive reconstruction

Where a decoder exists, reconstruct the point measure:

- D1: reconstruct the truncated density and perform species-resolved peak localization/refinement.
- D2: moment inversion / annihilating polynomial / multivariate Prony recovery followed by nonlinear refinement.
- D3: multidimensional Prony/annihilating-filter recovery from the selected Fourier sample set.

Report descriptor residual, assignment-aware Cartesian RMSD, pair-distance-matrix error, neighbor-count recovery, and species recovery.

5.3 Numerical inverse optimization

For randomly generated and real neighborhoods, solve from many random starts

$$\min_{\tilde{\mathcal{N}}} \left\| D(\tilde{\mathcal{N}}) - D(\mathcal{N}) \right\|^2. \quad (49)$$

Success is assessed after optimal species-preserving assignment. Failure to reconstruct does not prove incompleteness because the inverse optimization may fail. Conversely, discovery of two verified distinct configurations with descriptor discrepancy below numerical tolerance disproves injectivity at that truncation.

5.4 Adversarial collision search

Actively search for

$$\max_{\mathcal{N}, \mathcal{N}'} d_{\text{geom}}(\mathcal{N}, \mathcal{N}') \quad \text{subject to} \quad \|D(\mathcal{N}) - D(\mathcal{N}')\| \leq \varepsilon. \quad (50)$$

The test corpus must include same-radius shells, collinear/coplanar environments, high-symmetry polyhedra, chiral/reflected pairs, near-collisions, species exchanges, and atoms approaching the cutoff. Known invariant-descriptor counterexamples should be included as negative controls: the complete covariant descriptors should separate them while deliberately incomplete invariant baselines should not.

5.5 Jacobian rank and conditioning

Let $\mathbf{q} \in \mathbb{R}^{3N}$ be local Cartesian coordinates and

$$J_D = \frac{\partial D}{\partial \mathbf{q}}. \quad (51)$$

For a generic frame-aware descriptor, local injectivity requires $\text{rank} J_D = 3N$. An invariant descriptor modulo rotations has generic maximum $3N - 3$. After suitable nondimensionalization, inspect singular values

$$\sigma_1 \geq \cdots \geq \sigma_{3N}. \quad (52)$$

Unexpected exact null directions demonstrate local information loss; very small singular values indicate near-degeneracy and unstable inversion. High-symmetry structures with nontrivial stabilizers are analyzed separately rather than pooled into generic rank statistics.

5.6 Continuation along null and near-null directions

Starting from a right singular vector $v_{\min}$, perturb

$$\mathbf{q}_{k+1} = \mathbf{q}_k + \Delta t v_{\min} \quad (53)$$

and project back toward the descriptor level set. This converts infinitesimal conditioning information into a search for finite collisions and helps determine whether small singular values are benign local stiffness or evidence of extended ambiguous manifolds.

5.7 Stable completeness metrics

Report a local inverse-conditioning proxy

$$\kappa_{\text{inv}} \sim \sigma_{\min}(J_D)^{-1} \quad (54)$$

and an empirical collision-margin curve

$$\gamma_D(\delta) = \inf_{d_{\text{geom}}(\mathcal{N}, \mathcal{N}') \geq \delta} \|D(\mathcal{N}) - D(\mathcal{N}')\|. \quad (55)$$

Global numerical searches can only estimate γ_D , so the report must state it as an empirical lower/upper bound, not a proof.

5.8 Basis convergence

For each descriptor sweep the appropriate truncation variables: $(n_{\text{max}}, \ell_{\text{max}})$, moment/Zernike degree, frequency shells, ACE body degree and angular/radial limits. Record reconstruction, collision rate, Jacobian singular-value quantiles, descriptor dimension, bytes/atom, and precomputation throughput. Retain at least two practical resolutions per descriptor for downstream Hamiltonian experiments: a compact point on the stable-information curve and a high-resolution point near the storage/compute frontier.

Acceptance gate.

A finite descriptor configuration may enter the Hamiltonian screening grid only if: (i) exact symmetry tests pass; (ii) its metadata and basis are fully reproducible; (iii) all discovered collisions/near-degeneracies are characterized; (iv) reconstruction/Jacobian evidence is reported rather than hidden; and (v) storage/throughput fit the global budget. A descriptor need not be mathematically complete to proceed, but its completeness status must be explicit: proved on a stated domain, complete only in the basis limit, empirically stably reconstructive, or demonstrably incomplete.

6 Pairwise equivariant mapping architectures

Let x_i and x_j be either full or punctured endpoint descriptors and let b_{ij} denote fixed radial-spherical bond features. Every mapper in this section is a one-pair function. The comparison is designed to separate three sources of expressivity: information already exposed by the descriptor, legal angular generators created by the mapper, and nonlinear invariant coefficient functions.

6.1 M0: equivariant linear readout

The simplest map acts block-diagonally by irrep type:

$$\hat{h}_q^{Lp} = \sum_c W_{qc}^{Lp} x_c^{Lp}. \quad (56)$$

Only input channels with exactly the target (L, p) can contribute. M0 cannot create new parity or angular order and cannot form endpoint-bond or endpoint-endpoint correlations. Its purpose is diagnostic: when D4 already contains target covariants, M0 measures how much of the Hamiltonian problem has been moved into deterministic feature construction.

6.2 M1: fixed N-center / pair-ACE covariants

Build a deterministic bank of pair covariants such as

$$\Psi^{(1)} = [b^{\ell_b} \otimes x_i^{\ell_i}]_{Kp_K}, \quad (57)$$

$$\Psi^{(2)} = [x_i^{\ell_i} \otimes x_j^{\ell_j}]_{Kp_K}, \quad (58)$$

$$\Psi^{(3)} = [[x_i^{\ell_i} \otimes b^{\ell_b}]_K \otimes x_j^{\ell_j}]_{Lp}. \quad (59)$$

Construct the appropriate symmetric and antisymmetric sectors under $i \leftrightarrow j$ and use a linear or shallow equivariant readout. This is the closest native analogue of the N-center Hamiltonian representation and provides a controlled polynomial baseline with no learned hidden equivariant state.

6.3 M2: dense global $O(3)$ Clebsch-Gordan E3MLP

Concatenate all pair input irreps, then alternate equivariant linear maps, full legal tensor products, and equivariant nonlinearities:

$$x^{(t+1)} = x^{(t)} + \mathcal{L}_2 \mathcal{A} \left([\mathcal{L}_a x^{(t)} \otimes \mathcal{L}_b x^{(t)}]_{\mathcal{I}_t} \right). \quad (60)$$

$\mathcal{A}$ may be NormAct, a $0e$ scalar gate, a strictly invariant FiLM modulation, or another proven equivariant activation. M2 retains all selected global CG paths and is the high-cost accuracy reference.

Strict implementation rules are:

- unrestricted scalar nonlinearities and biases only on $0e$;
- $0o$ scalar nonlinearities must be odd;
- gate values multiplying nonscalar irreps must be $0e$;
- invariant FiLM may use $0e$ channels, norms, and even contractions, but not raw $0o$ values in an unrestricted scalar MLP;
- residual or LayerScale parameters must be one scalar per irrep copy and broadcast over all $2\ell + 1$ magnetic components;
- dropout or normalization must operate on complete irrep copies, never arbitrary magnetic coordinates.

6.4 M3: invariant reduced-coefficient hypernetwork

This is the main conceptual candidate. Construct a fixed bank of equivariant generators

$$\Psi_{\alpha q}^{Lp} = \Psi_{\alpha q}^{Lp}(x_i, b_{ij}, x_j) \quad (61)$$

and a bank of true invariants

$$I_\beta = I_\beta(x_i, b_{ij}, x_j) \in 0e. \quad (62)$$

A conventional scalar MLP produces reduced coefficients

$$a_{q\alpha} = f_{q\alpha, \theta}(I_1, \dots, I_K, r_{ij}, Z_i, Z_j), \quad (63)$$

and the full target is

$$\hat{h}_q^{Lp} = \sum_{\alpha} a_{q\alpha} \Psi_{\alpha q}^{Lp}. \quad (64)$$

Proposition 2. M3 is exactly $O(3)$ -equivariant if every $a_{q\alpha}$ is $0e$ and every generator transforms in the declared target irrep.

Reason. For $Q \in O(3)$, invariance gives $a_{q\alpha}(Qz) = a_{q\alpha}(z)$, while $\Psi_{\alpha}(Qz) = D^{Lp}(Q)\Psi_{\alpha}(z)$. Hence

$$\hat{h}(Qz) = \sum_{\alpha} a_{\alpha}(z) D(Q) \Psi_{\alpha}(z) = D(Q) \hat{h}(z). \quad \square \quad (65)$$

This architecture is a neural Wigner-Eckart construction: fixed angular structures carry covariance and an unrestricted scalar network learns environment-dependent reduced coefficients. Its key approximation is generator completeness, which can be systematically enlarged by polynomial degree, intermediate angular momentum, radial degree, and endpoint-exchange sector.

6.5 M4: explicit trilinear coupling-tree network

Treat (x_i, b_{ij}, x_j) as three typed inputs and compare recoupling orders

$$\Psi_A^L = [[x_i \otimes b]_K \otimes x_j]_L, \quad (66)$$

$$\Psi_B^L = [[x_i \otimes x_j]_{K'} \otimes b]_L, \quad (67)$$

$$\Psi_C^L = [x_i \otimes [x_j \otimes b]_{K''}]_L. \quad (68)$$

If all intermediate irreps are retained these coupling trees are related by Wigner-6j recoupling. Under truncation they become distinct inductive biases. The implementation should exploit this by precomputing endpoint-only products when possible and delaying the pair-specific bond coupling.

6.6 M5: factorized or depthwise tensor products

A fully connected multiplicity-space tensor product can have path weights $W_{c_i c_j c_o}^{(\pi)}$. Factorize them, e.g.

$$W_{c_i c_j c_o}^{(\pi)} \approx \sum_{r=1}^R U_{c_i r}^{(\pi)} V_{c_j r}^{(\pi)} O_{r c_o}^{(\pi)}, \quad (69)$$

or use MANDALA-style separate weights/depthwise angular products followed by same-irrep channel mixing. The CG coefficients and legal paths remain exact; only multiplicity mixing is restricted. Sweep rank R or channel multiplier at matched compute.

6.7 M6: learned sparse tensor-product paths

Enumerate a complete candidate path set

$$\pi = (\ell_i, p_i, \ell_j, p_j, \ell_b, p_b, K, L, p_L, \text{radial indices}) \quad (70)$$

and assign a path-level invariant gate g_π . The output is

$$\hat{h}^{Lp} = \sum_{\pi \rightarrow Lp} g_\pi \Psi_\pi^{Lp}. \quad (71)$$

Gates act on entire irrep paths, never individual m components. The planned schedule is dense warm-up, gradual gate sharpening/sparsity regularization, structural pruning, and final fine-tuning. SPHNet motivates the efficiency idea; its graph architecture is not reproduced.

6.8 M7: complete local-O(2) bond-frame mapper

M7 rotates endpoint irreps to a bond-aligned frame, performs a parity-complete O(2)-equivariant network there, and rotates target irreps back. It is treated in detail in Section 7 because its exactness conditions are subtle. This is the primary efficiency hypothesis.

6.9 M8: Gaunt/Fourier spherical-function mapper

Interpret angular coefficients as a band-limited spherical function

$$f(\hat{n}) = \sum_{\ell m} x_{\ell m} Y_{\ell m}(\hat{n}), \quad (72)$$

use Gaunt products or Fourier-basis convolutions to realize selected angular products, and transform back. The appeal is asymptotic angular scaling closer to $O(L^3)$ rather than dense $O(L^6)$ products. Parity is a hard requirement: a plain natural-parity scalar spherical field does not by itself span simultaneous polar and pseudotensor target channels. M8 must either use explicitly parity-labelled doubled fields or fall back to global/O(2) operations for pseudo sectors.

6.10 M9: symmetry-adapted kernel regression

Use deterministic pair covariants with linear or symmetry-adapted nonlinear kernels. This is the principal non-neural control after ACE: it separates descriptor and angular-feature quality from neural optimization. The N-center linear/KRR implementation should share the same full-block irrep schema, split, R_H , envelope, and headline matrix-element metrics.

6.11 Mapper comparison matrix

Mapper	Main idea	Creates new irreps	Learned nonlinear scalar map	Expected cost
M0	equivariant linear readout	no	no	very low
M1	fixed pair/N-center covariants	yes, fixed	optional/shallow	low-medium
M2	dense global CG E3MLP	yes	yes	high
M3	invariant coefficients $\times$ fixed generators	yes, fixed bank	yes	medium
M4	explicit trilinear trees	yes	optional	medium-high
M5	factorized/depthwise TP	yes	yes	medium
M6	learned sparse TP paths	yes	yes	adaptive
M7	bond-frame complete $O(2)$	yes	yes	low-medium at high L
M8	Gaunt/Fourier TP	yes	yes	favorable at high L
M9	symmetry-adapted kernel	via feature bank	kernel	training/memory dependent

7 Fast bond-frame tensor products and complete $O(2)$ processing

7.1 Axis alignment

For a nonzero pair define $\hat{n} = \mathbf{r}_{ij}/r_{ij}$ and select a rotation $R_{\hat{n}}$ such that

$$R_{\hat{n}}\hat{n} = \mathbf{e}_z. \quad (73)$$

Rotate every endpoint irrep:

$$\tilde{x}_i^{\ell p} = D^{\ell p}(R_{\hat{n}})x_i^{\ell p}, \quad \tilde{x}_j^{\ell p} = D^{\ell p}(R_{\hat{n}})x_j^{\ell p}. \quad (74)$$

In this frame

$$Y_{\ell m}(\mathbf{e}_z) = 0 \quad (m \neq 0), \quad (75)$$

which dramatically sparsifies bond-harmonic tensor products.

7.2 Residual gauge and the stabilizer group

$R_{\hat{n}}$ is not unique. Any $R_z(\phi)R_{\hat{n}}$ also sends the bond to $\mathbf{e}_z$. Therefore the internal bond-frame model may not be an arbitrary MLP: it must be equivariant under the stabilizer of the axis.

For proper rotations the stabilizer is $SO(2)$. In a complex basis,

$$\tilde{x}_m \mapsto e^{im\phi}\tilde{x}_m, \quad (76)$$

so products conserve local azimuthal charge, $m_{\text{out}} = m_1 + m_2$. In a real basis the two-dimensional sine/cosine irreps produce sum/difference orders.

For full $O(3)$, the stabilizer is

$$\{g \in O(3) : g\mathbf{e}_z = \mathbf{e}_z\} \simeq O(2), \quad (77)$$

not merely $SO(2)$. This distinction is essential for Hamiltonians because target blocks contain both polar and pseudo irreps.

7.3 Why $SO(2)$ -only is insufficient

A p - p Hamiltonian sector contains $1e$, a pseudovector. Local $SO(2)$ alone cannot distinguish all global polar/pseudo channels that restrict to similar azimuthal behavior. A complete implementation must retain parity/reflection information and the parent global (ℓ, p) labels when required. In particular:

- local $m = 0$ has distinct even and odd one-dimensional $O(2)$ representations;
- only the even scalar may receive arbitrary bias;
- odd local scalars require odd activation;
- local tensor products must track reflection parity as well as azimuthal order;

- mapping back to global irreps must know the parent (ℓ, p) target channel and multiplicity.

The recent full- $O(3)$ local- $O(2)$ construction is directly relevant but must be independently validated because it is a new 2026 result.

7.4 Global equivariance theorem for a bond-frame map

Let T_n rotate a global representation into the chosen bond frame and let Φ be equivariant under the full local stabilizer $H \simeq O(2)$. Define

$$\mathcal{F}_n(x) = T_n^{-1} \Phi(T_n x). \quad (78)$$

Proposition 3. If the frame-selection convention is measurable/continuous away from its unavoidable chart singularities and Φ is H -equivariant, then $\mathcal{F}$ is independent of the arbitrary roll choice and is globally $O(3)$ -equivariant after the standard gauge-transition relation is enforced.

Reason. Two valid frames differ by $h \in H$. Replacing T by hT changes the internal input by h and, by H -equivariance, changes the internal output by the corresponding h . The inverse frame rotation cancels that change. Under a global Q , the aligned frames differ by a stabilizer element, so the same cancellation yields $\mathcal{F}(Qx) = D_{\text{out}}(Q)\mathcal{F}(x)$. $\square$

The implementation will test this numerically using multiple independent frame constructors and random roll perturbations; identical global outputs must result.

7.5 Three notions of “fast tensor product”

The paper and code must distinguish:

1. **Exact reduction of node-irrep $\otimes$ bond harmonic.** Since the aligned bond harmonic is $m = 0$, many CG terms vanish exactly. This is the eSCN-style source of large savings.
2. **A general local $SO(2)$ pair network.** It can be globally $SO(3)$ -equivariant with a valid gauge construction, but it need not be algebraically identical to a chosen global CG network.
3. **A parity-complete local $O(2)$ operator system.** This is the required production target for full spinless Hamiltonian irreps because pseudo channels coexist with polar channels.

Claims of exact equivalence must state which of these is meant.

7.6 M7 production design

For offsite pairs:

1. rotate D_i, D_j into the bond frame;
2. decompose/reindex global (ℓ, p) copies into local $O(2)$ channels while retaining parent metadata;
3. create selected deterministic local degree-2/3 covariants;
4. create true local $0e$ invariants from scalar channels, norms, scalar products, legal contractions, r_{ij} , species and pair metadata;
5. feed invariants to an ordinary MLP to predict reduced coefficients;
6. form target local covariants and optionally apply one or two residual O2Gate blocks;
7. map local target channels back to the declared complete global block-irrep vector;
8. rotate back to the global frame;
9. apply exact pair-exchange/Hermiticity construction;
10. multiply by the frozen range envelope.

Onsite blocks use a separate global- $O(3)$ reduced-coefficient mapper because there is no nonzero bond axis.

Acceptance gate.

M7 cannot enter benchmarking until it passes four independent properties in float64: global proper-rotation equivariance, inversion/improper-reflection equivariance, arbitrary bond-frame roll invariance, and exact agreement of local-vs-global tensor products on a controlled exhaustive set of low- ℓ coupling paths. A merely $SO(3)$ -equivariant implementation is insufficient.

8 Range-magnitude factorization, loss, and normalization

8.1 Frozen range-magnitude transform

For each dataset define a fixed diagonal transform in full-block irrep space,

$$G_{ij}(r) = \bigoplus_q g_q(r, Z_i, Z_j) I_{2L_q+1}, \quad (79)$$

where one scalar g_q multiplies every magnetic component of irrep copy q . The learned network predicts a normalized residual vector $\tilde{\mathbf{h}}$ and the physical prediction is

$$\hat{\mathbf{h}} = G(r)\tilde{\mathbf{h}}. \quad (80)$$

g_q is $0e$, positive where necessary, compatible with pair reversal, and fitted only from the training partition. A single envelope-building algorithm and frozen fitted parameters are shared by every model on a dataset.

Permitted envelope families include the existing MANDALA Slater-like soft-cutoff constructions and any later replacement only if the replacement is frozen before comparing descriptors/mappers. The study does not allow each architecture to tune a separate envelope.

8.2 Primary irrep-space objective

The default physical loss is computed after range restoration:

$$\mathcal{L}_{\text{irrep}} = \frac{1}{N_{\text{coord}}} \sum_{(i,j)} \sum_q w_{ijq} \left\| G_q(r_{ij}) \tilde{h}_{ij,q} - h_{ij,q} \right\|_2^2. \quad (81)$$

If the MANDALA block-to-irrep transform is orthonormal under vectorization, the unweighted coordinate sum is identical to physical block Frobenius error:

$$\left\| \hat{H} - H \right\|_F^2 = \left\| \hat{\mathbf{h}} - \mathbf{h} \right\|_2^2. \quad (82)$$

This equality is a required unit test. If the transform is not orthonormal, derive and use the induced metric G_Q in the loss rather than silently changing the physical weighting.

Initial experiments use natural occurrence frequencies. Natural-frequency or inverse-frequency reweighting is introduced only when a documented imbalance causes rare pair/irrep sectors to be systematically ignored. Every weighted result must also report the unweighted physical metrics.

8.3 Irrep-space versus matrix-space loss ablation

The main project trains on irrep vectors. A later controlled ablation compares:

$$\mathcal{L}_{\text{irrep}} = \left\| \Delta \mathbf{h} \right\|_{G_Q}^2, \quad (83)$$

$$\mathcal{L}_{\text{matrix}} = \left\| \Delta H \right\|_F^2. \quad (84)$$

When the transform is exactly orthonormal these objectives should agree mathematically, and any performance/runtime difference arises from implementation, numerical conditioning, batching, or added weighting. The ablation is therefore a consistency/performance test, not a central scientific hypothesis.

8.4 Descriptor and hidden-feature normalization

All normalization statistics are computed on the training partition. For an irrep copy (L, p, q) , scaling must be identical for all $2L + 1$ magnetic components. Acceptable statistics include RMS norm per irrep copy or a shared scale over a declared multiplicity group. Arbitrary per- m standardization is prohibited. Additive centering is permitted for $0e$ scalars; it is not permitted for nonscalar irreps and must preserve odd parity for $0o$.

The normalized residual target h/g will be monitored separately because a physically sensible envelope should reduce dynamic range without amplifying noise where g is tiny. Define an envelope floor or mask only from the training-data range support, and document its effect on the physical target.

8.5 Headline metrics

Every experiment reports at least:

- MAE over all reconstructed physical Hamiltonian matrix elements in meV – the headline metric;
- matrix-element RMSE;
- complete-block Frobenius MAE/RMSE;
- error by species pair, distance bin, onsite/offsite class, target (L, p) irrep and multiplicity;
- Hermiticity residual before/after exact projection if applicable;
- equivariance residual;
- parameter count, descriptor bytes/atom, pair throughput, end-to-end wall time, and peak accelerator memory.

For final models additionally report band/eigenvalue and density-of-states errors under a fixed reciprocal-space evaluation protocol.

9 Benchmark datasets and baseline hierarchy

9.1 Primary public dataset: HamGNN SiO₂

The main dataset is the SiO₂ benchmark released with HamGNN. It contains 221 Materials Project SiO₂ crystal structures, each perturbed three times by random 0.1 Å displacements, giving 663 structures with cells up to 80 atoms. Hamiltonians were generated with OpenMX. HamGNN used an 80/10/10 random split and reports a Hamiltonian matrix-element MAE of 2.29 meV on its SiO₂ test set. A larger 180-atom SiO₂ transfer example is reported at 1.67 meV.

9.1.1 Split protocol

Stage 1 first attempts to recover the exact published split indices from the released artifacts/code. If that is impossible, reproduce the published *protocol*: random structure-level 80/10/10 assignment with a frozen project seed. Such numbers must be labelled “protocol-matched, split-not-identical” and the 2.29 meV HamGNN result remains contextual rather than an exact head-to-head comparison.

At the end of the project, add a parent-polymorph-grouped split in which all three perturbations derived from one Materials Project parent remain in the same partition. This grouped split is expected to be harder and scientifically cleaner; it is intentionally deferred so the first development cycle stays comparable to the published HamGNN protocol.

9.2 Internal 150-atom bulk SiO₂

The internal 150-atom bulk SiO₂ dataset is reserved for same-chemistry large-cell validation. If its DFT functional, pseudopotentials, AO basis, orbital ordering and gauge can be made exactly compatible with the public benchmark, evaluate zero-shot models trained on public SiO₂. If not, treat it as a separate internal benchmark and do not mix targets across incompatible AO conventions.

This dataset tests the core locality hypothesis: a fixed local descriptor + pair map should have cost and accuracy that are largely independent of total cell size once neighborhoods are statistically in distribution.

9.3 Secondary public chemistry: monolayer MoS₂

The final public cross-chemistry benchmark is the 5×5 monolayer MoS₂ dataset used in the DeepH-E3/SLEM benchmark family: 500 AIMD structures with 75 atoms and OpenMX Hamiltonians. It provides a binary chemistry with Mo d orbitals and therefore richer high- L Hamiltonian sectors than SiO₂.

Published message-passing numbers are contextual only. Representative reported MoS₂ Hamiltonian-element MAEs include approximately 0.34 meV for a compact SLEM model, 0.14 meV for a larger SLEM model, 0.46–0.55 meV for DeepH-E3 variants, and 1.20 meV for HamGNN. The project will not attempt to reproduce these GNNs. Only the best reduced subset of descriptor/mapper combinations will be transferred to MoS₂.

9.4 Internal debugging datasets

The perturbed 8-atom periodic Si dataset is used for deterministic CI, exhaustive rotation/reflection tests, finite-difference Jacobians, and rapid overfit/sanity checks. It is not a headline benchmark because it is unary and structurally narrow. The 32-atom ZnCuSn(SeS) dataset is deferred because the first paper is restricted to two- or three-element systems.

9.5 Stage-1 reproduced baseline: equivariant ACE Hamiltonian

The native baseline follows the ACEhamiltonians approach: a deterministic atom-centered equivariant polynomial basis and linear Hamiltonian regression, with separate mathematical treatment of onsite and offsite environments but one full-block MANDALA target interface. The official ACEhamiltonians package is used as a reference oracle only; production experiments use a native implementation integrated with exactly the same dataset, block mapper, envelope, loss, logging, and metrics as all new methods.

The original ACEhamiltonians publication demonstrates non-message-passing Hamiltonian learning for crystalline Al. Its reported onsite model uses higher correlation/body order than the offsite model; offsite low-order features were sufficient in that benchmark. Published Al errors are used only to validate the order of magnitude and implementation behavior, never as numerical competitors to SiO₂.

9.6 Second reproduced non-GNN baseline: N-center equivariant regression

After Stage 1 infrastructure is stable, implement the Nigam-Willatt-Ceriotti N-center Hamiltonian representation using a fixed pair-centered equivariant feature bank and symmetry-adapted linear or kernel regression. It is highly relevant because it explicitly treats two tagged atomic centers, spatial covariance, parity, and exchange symmetry without graph message passing.

9.7 Published GNN references

HamGNN, DeepH-E3, SLEM, MACE-H, and related message-passing models are literature reference points only. The paper must not imply controlled reproduction where none was performed. Comparison tables must label each number as one of:

- **native reproduced** – same project pipeline;
- **upstream reference reproduced** – official package/run on controlled input;
- **published contextual** – copied from literature with dataset/split caveats.

9.8 Determining the Hamiltonian cutoff R_H

R_H is not a tunable neural hyperparameter. For each dataset it is frozen before model training using training structures only. Build distance-binned distributions of block Frobenius norm, maximum element magnitude, and aggregate omitted-tail contribution. Select the smallest R_H satisfying both:

1. the estimated mean matrix-element error floor from omitting all blocks beyond R_H is at most 0.1 meV or 5% of the target benchmark MAE, whichever is stricter;
2. at least 99.9% of the observed offsite target squared Frobenius mass is retained, unless the AO basis has an exact documented shorter support.

If no practical radius satisfies this, the dataset does not support the intended sparse local block protocol without revising the target/envelope assumptions. The exact numeric thresholds may be tightened before Stage 1 but, once frozen, must be shared by every method.

9.9 Descriptor-radius sweep

R_D is independent of R_H and may be larger. Construct a data-informed sweep with at least four points, normally

$$R_D \in \{R_H, R_H + 1 \text{ \AA}, R_H + 2 \text{ \AA}, R_H + 4 \text{ \AA}\}, \quad (85)$$

subject to periodic geometry and memory. Add a smaller-radius negative control if informative. Do not select R_D by test error; select candidate radii using validation and Stage-6 locality tests, then evaluate the frozen choice once on test.

10 Research hypotheses and expected outcomes

The project is hypothesis-driven. “Expected” below is a prior to be tested, not a conclusion. Every stage report must mark each applicable hypothesis as supported, contradicted, unresolved, or not tested.

ID	Falsifiable hypothesis	Expected outcome / interpretation
H1	A practical truncation of D1 can be stably reconstructive for the SiO ₂ neighborhood distribution even though exact completeness is only guaranteed in the complete-basis limit.	High reconstruction success; residual near-collisions concentrated at cutoff/high-symmetry cases.
H2	D2 admits the strongest finite-domain injectivity statement among the four descriptors.	A theorem or clearly delimited generic/stable recovery result under bounded N and separation assumptions.
H3	D3 offers better reconstruction conditioning per stored coefficient than raw high-degree moments.	Larger lower-tail Jacobian singular values at matched bytes, but possible aliasing at poor frequency grids.
H4	D4 improves finite-data Hamiltonian sample efficiency relative to D1 at matched mapper depth because useful high-order/pseudo covariants are exposed deterministically.	Lower error at small/medium training sizes; diminishing advantage with very expressive mappers.
H5	Pair puncturing improves offsite generalization by separating the explicitly known partner from the surrounding environment.	Modest gain or no harm; if worse, duplication was serving a useful interaction channel.
H6	M3 reduced-coefficient models approach dense M2 accuracy at substantially lower compute because the difficult nonlinear function is mostly invariant.	Best accuracy/cost global-frame candidate.
H7	Full local-O(2) M7 matches global M2 accuracy within statistical uncertainty while reducing high- L tensor-product cost.	Strongest Pareto candidate, especially on MoS ₂ .
H8	A merely local-SO(2) implementation exhibits parity failures or reduced accuracy on pseudo-irrep sectors; full O(2) removes them.	Directly visible on $1e/2o/\dots$ unit tests and target-sector errors.
H9	Factorized M5 and sparse M6 retain most of M2 accuracy at lower parameter/FLOP cost.	Useful mid-Pareto points; path sparsity may be target/descriptor dependent.
H10	Increasing R_D improves accuracy only until target sufficiency saturates; beyond that point storage and compute increase without material benefit.	Plateau identifiable by validation error and outside-cutoff response.
H11	Descriptor reconstruction/Jacobian quality predicts Hamiltonian performance only until the target-sufficiency limit; beyond that, extra geometric information has diminishing value.	Correlation across coarse descriptors, weak correlation among already sufficient variants.
H12	The frozen range-magnitude factorization improves conditioning and convergence without materially reducing achievable physical accuracy.	Faster convergence and lower gradient dynamic range; no significant test penalty.
H13	Irrep-space and matrix-space physical losses are equivalent when the block mapper is orthonormal; measured differences are implementation/weighting effects.	Near-identical optima under matched physical weighting.
H14	A local pair model trained on public SiO ₂ transfers to much larger same-chemistry cells without a systematic cell-size error increase when local environments are covered.	Internal 150-atom result near the in-distribution local-environment error, modulo OOD structures.

ID	Falsifiable hypothesis	Expected outcome / interpretation
H15	The parent-grouped SiO ₂ split is harder than the random perturbation split but preserves the qualitative ranking of descriptor/mapper families.	Higher errors; ranking mostly stable.
H16	Nonlinear pair maps outperform the native linear ACE/N-center baselines once descriptor and target cutoffs are held fixed, without requiring message passing.	Main empirical claim if supported.

11 MANDALA implementation architecture

11.1 Data products and immutable schemas

All cached artifacts are keyed by a content hash of: dataset version, structure identity, descriptor class/version, R_D , basis parameters, species ordering, real-harmonic convention, periodic-image convention, numeric dtype, and software commit. A cache entry lacking this metadata is invalid.

Each dataset produces the following immutable tables/shards:

1. structure metadata and split assignment;
2. canonical atom and periodic-pair lists up to R_H ;
3. exact/ragged neighbor sets up to the largest planned R_D where storage permits;
4. one cache per descriptor/resolution with declared `irreps_out`;
5. target full-block irrep vectors and AO reconstruction metadata;
6. frozen range-envelope parameters;
7. train-only normalization statistics;
8. optional analysis caches: descriptor Jacobians, reconstruction results, and DFT intervention pairs.

11.2 Descriptor API

Every production descriptor implements the conceptual interface

```
class AtomicDescriptor:
    cutoff: float
    irreps_out: Irreps
    species_order: tuple
    basis_metadata: dict

    compute(structure, neighbor_set) -> PerAtomIrrepArray
    contribution(species, displacement, image) -> IrrepArray
    jacobian(environment) -> Array # analytic/autodiff
    reconstruct(descriptor, metadata) -> Neighborhood # where available
    certify_metadata() -> dict
```

`contribution` is mandatory for additive descriptors if puncturing is claimed. It must exactly reproduce the term included by `compute` for the same neighbor/image.

11.3 Full-block pair-mapper API

All learned and linear models share one public interface:

```
class PairHamiltonianMapper:
    input_irreps: Irreps
    target_block_irreps(pair_type) -> Irreps
```

```

forward(
    descriptor_i,
    descriptor_j,
    displacement_ij,
    species_i,
    species_j,
    pair_type_metadata,
) -> full_block_irrep_vector

```

The returned vector has the exact canonical MANDALA target schema for that full AO block. The implementation may batch different pair types by padding, ragged grouping, or type-specific modules, but it may not expose shell-pair prediction as the scientific model interface.

11.4 No-message-passing code audit

A static/runtime audit is part of model acceptance. During one pair forward pass, the model may not read descriptors of any atom other than i and j , except immutable dataset-level constants. The pair mapper code must not contain learned scatter/reduce over graph edges, node-state caches updated by other pairs, or iterative graph propagation. Deterministic descriptor precomputation is outside the learned forward graph.

A unit test constructs two structures with identical pair inputs ($D_i, D_j, \mathbf{r}_{ij}$) but different irrelevant graph ordering/connectivity metadata; output must be bitwise identical in deterministic mode.

11.5 Target schema and multiplicity bookkeeping

For each species pair, enumerate every AO basis function in the canonical OpenMX order. Construct shell-pair CG transforms internally, concatenate their irreps, and then define one immutable full-block order. Store for every irrep copy q :

- target (L, p) ;
- multiplicity index in the full block;
- source AO row/column ranges;
- shell/radial labels for diagnostics only;
- reverse-pair mapping index and phase/sign matrix;
- range-envelope channel index.

Round-trip and pair-reversal transforms are generated from this metadata, not from hand-maintained tables.

11.6 Precomputation and 2 TB storage policy

The 2 TB project budget is a hard aggregate limit including descriptors, exact neighbor sets, target caches, model checkpoints, and analysis artifacts. The precompute pipeline must estimate final size before writing a new cache. Normal operation reserves at least 10% headroom, so automatic descriptor-cache generation stops if projected persistent usage exceeds 1.8 TB without explicit project review.

Recommended formats are chunked, sharded arrays with per-shard checksums and atomic completion markers. Exact implementation may use Zarr, HDF5, or a MANDALA-native equivalent; the scientific requirements are random pair/atom access, efficient sequential training reads, metadata integrity, and resumable generation.

Production descriptors are normally stored in float32 after float64 generation/verification. A representative float64 validation shard is retained for symmetry and reconstruction regression tests. Ragged neighbor sets store compact displacement arrays, integer species IDs, periodic image vectors, and prefix offsets. Compression must be lossless for metadata and neighbor identity.

11.7 Reproducibility and experiment registry

Every run records:

- git commit and uncommitted-diff hash;
- dataset/cache/split/envelope hashes;

- complete YAML/JSON configuration;
- random seeds and deterministic-library flags;
- hardware/software versions;
- parameter count and declared theoretical FLOP estimator version;
- wall-clock preprocessing/training/evaluation timings;
- final and best-validation checkpoints;
- machine-readable metrics by global and stratified categories.

No table in the final paper may contain a manually copied experimental number without a traceable registry record, except explicitly labelled published literature values.

12 Experimental protocol and comparison policy

12.1 Training seeds and model selection

For learned neural pair mappers, use at least three independent training seeds for final comparisons. Broad screening may use one seed to remove clearly dominated configurations, but any promoted configuration is rerun with three seeds before a scientific conclusion. Deterministic linear baselines require only one fit unless stochastic numerical solvers are used.

Hyperparameters are selected from validation only. Test evaluation is performed after the configuration is frozen. The final grouped-SiO₂ and internal-OOD evaluations are treated as confirmatory tests; no architecture may be redesigned after observing them without labelling the new result exploratory.

12.2 Fair resource comparisons

Report both unrestricted-best and matched-resource comparisons. At minimum define three pair-mapper budget bands by parameter count and measured pair-forward cost. When feasible compare methods at approximately matched:

- learnable parameters;
- training/inference pair FLOPs or measured throughput;
- peak accelerator memory.

Descriptor methods must additionally be compared at matched stored bytes/atom and at matched reconstruction quality. A method may win on accuracy while losing on storage; the paper must show the Pareto frontier rather than compressing all costs into one score.

12.3 Learning curves

For the primary SiO₂ benchmark, train promoted methods at several fractions of the training set, e.g. 10%, 25%, 50%, 75%, 100%, with parent/structure sampling performed before pair expansion. Fit learning-curve slopes only when the same validation/test protocol and optimizer budget are maintained. The purpose is to determine whether deterministic high-order descriptors primarily improve sample efficiency or also improve the asymptotic error floor.

12.4 Statistical conclusions

For stochastic models report mean, standard deviation, and individual seed values. A ranking is called robust only when it is reproduced across at least two seeds without an obvious compute-budget confound and the mean difference is larger than the pooled seed noise. For close methods, use paired bootstrap resampling over test structures/blocks to report confidence intervals on matrix-element MAE differences.

12.5 Mandatory stratification

Global MAE can hide serious physical failures. Every final comparison includes error against:

- pair distance and envelope magnitude;

- onsite versus offsite;
- species pair and directed/canonical pair type;
- target L and parity p ;
- AO shell/radial source sector after reconstruction;
- block Frobenius magnitude;
- coordination/environment descriptors;
- structure size and parent polymorph where available.

When an architecture is parity-incomplete, pseudo-irrep errors must be shown explicitly rather than averaged with easier natural-parity sectors.

13 Staged execution plan

Every stage ends with a concise but exhaustive report using the template in Section 14. A critical acceptance failure blocks progression or triggers a documented scope change. “Expected outcome” is a pre-registered prediction, not a success criterion.

13.1 Stage 1 - Native non-GNN baseline and benchmark plumbing

Purpose. Establish one trustworthy non-message-passing baseline and a common evaluation pipeline before developing new descriptors or mappers.

Hypotheses tested. Infrastructure only; establishes reference points for H12/H16 later.

Work items.

1. Ingest HamGNN SiO₂, canonicalize OpenMX AO ordering, and precompute full-block target irreps.
2. Attempt recovery of the exact HamGNN random split; otherwise freeze a protocol-matched 80/10/10 split and record the distinction.
3. Establish R_H from training data using the omitted-tail protocol.
4. Fit and freeze one dataset-wide MANDALA range-envelope construction.
5. Implement the full-block AO $\leftrightarrow$ irrep round trip, pair-reversal map, physical metrics, and stratified evaluation.
6. Install/run upstream ACEhamiltonians on its controlled/released AI examples; record features/predictions where accessible.
7. Implement a native ACEhamiltonians-style equivariant ACE basis/regressor inside the common MANDALA workflow.
8. Build explicit basis-index/normalization mapping between upstream and native ACE representations and compare on controlled AI structures.
9. Train/evaluate the native ACE baseline on the frozen SiO₂ split and save its complete performance/cost profile.

Deliverables.

- frozen SiO₂ split, pair lists, R_H , target-irrep cache, range envelope, and metric implementation;
- native ACE baseline code/config/checkpoint;
- upstream-vs-native ACE comparison artifact;
- Stage-1 benchmark report.

Acceptance gate.

Stage 1 passes only if all of the following are satisfied:

1. AO $\rightarrow$ irrep $\rightarrow$ AO round-trip relative Frobenius error $\leq 10^{-10}$ in float64 on random full blocks, and $\leq 10^{-5}$ in float32.

2. Analytic pair reversal in irrep space reproduces AO transpose/Hermiticity with relative error $\leq 10^{-10}$ float64.
3. $O(3)$ transformation tests of the target mapper are $\leq 10^{-9}$ relative float64 for well-scaled random blocks; all convention-dependent outliers are resolved.
4. After explicit index/normalization alignment, native and upstream ACE basis values agree to max relative error $\leq 10^{-8}$ and max absolute error $\leq 10^{-10}$ on a controlled float64 test set where both implementations represent the same basis. If the upstream package does not expose identical basis values, compare reconstructed physical predictions under shared coefficients to $\leq 10^{-6}$ eV per matrix element and document the limitation.
5. The upstream/native AI baseline reproduces the published qualitative convergence and error scale; any numerical mismatch is explained by dataset/basis/split differences rather than ignored.
6. The SiO_2 ACE baseline completes end-to-end with no data leakage, reports physical matrix-element MAE, and has a fully reproducible run record. There is no required SiO_2 accuracy threshold at this stage: the value becomes the project baseline.

Expected outcome. Native ACE should agree nearly exactly with the reference where mathematical basis definitions coincide. On SiO_2 it is expected to underperform modern message-passing published numbers but establish a strong, interpretable non-GNN floor.

13.2 Stage 2 - Descriptor infrastructure and large-scale precomputation

Purpose. Implement D1-D4 and the point-set oracle with exact irrep metadata, caching, puncturing, and storage controls.

Work items.

1. Implement D1 with at least two orthogonal radial basis choices.
2. Implement D2 Zernike/STF moments and analytic Jacobians.
3. Implement D3 Fourier-Bessel descriptor with configurable frequency grid.
4. Implement deterministic D4 degree-2 and selected degree-3 lifts.
5. Store ragged exact neighbor sets to the largest practical R_D .
6. Implement common descriptor normalization, cache hashing, sharding, and size estimator.
7. Generate a data-informed R_D sweep and at least two resolution levels per descriptor.
8. Implement exact puncturing consistency tests for D1/D2/D3 where additive.

Acceptance gate.

Stage 2 passes when every descriptor: (i) has a declared exact `irreps_out`; (ii) passes random rotation, inversion/reflection, equal-species permutation, PBC remapping and deterministic-recompute tests at the pre-registered numeric tolerances; (iii) has no unexplained NaN/overflow at the largest planned basis; (iv) has measured precompute throughput and bytes/atom; and (v) the projected project footprint remains below 1.8 TB before reserve. Additive descriptors must reproduce full-minus-single-neighbor puncturing to $\leq 10^{-10}$ relative float64 on controlled cases.

Expected outcome. D1 and D3 should be straightforward and stable at moderate order; D2 high-order raw forms will motivate orthogonal scaling; D4 storage will grow fastest and require deterministic path caps.

13.3 Stage 3 - Completeness theorem and empirical certification

Purpose. Establish what information each finite descriptor actually retains before using Hamiltonian accuracy as evidence.

Work items.

1. Write a formal D1 complete-basis injectivity argument with cutoff/species assumptions.
2. Attempt a finite-domain injectivity/recovery theorem for D2; if D3 obtains a stronger sampling theorem during development, present both.
3. Implement analytic/constructive reconstruction where possible.

4. Run numerical inverse optimization, adversarial collision search, Jacobian-rank analysis and null-direction continuation on synthetic bounded- N domains and sampled SiO_2 neighborhoods.
5. Create known-degeneracy negative controls for incomplete invariant representations.
6. Produce basis-convergence curves versus dimension, bytes, and runtime.
7. Select compact and high-information descriptor resolutions for Stage 5 without using test Hamiltonian error.

Acceptance gate.

A theorem is accepted only with explicit domain assumptions and a checkable proof/reduction to a cited recovery theorem. For empirical certification, on synthetic generic neighborhoods inside the declared domain the preferred resolution should reconstruct at least 99% of cases to species-aware Cartesian RMSD $< 10^{-4}$ Å from a robust multi-start procedure, or the report must document why this target is impossible/ill-conditioned. Unexpected Jacobian rank deficiency may occur in at most 0.1% of generic test environments after numerical-threshold review; all systematic failures and high-symmetry exceptions must be characterized. These percentages are acceptance targets for the selected practical resolution, not definitions of mathematical completeness.

Expected outcome. D2 should support the clearest finite-domain theorem. D1/D3 should demonstrate strong empirical stable reconstruction at practical resolution. D4 should inherit D1 completeness while improving availability of high-order/pseudo covariants.

13.4 Stage 4 - Pair-mapper correctness suite

Purpose. Implement and validate the pair maps before large training sweeps.

Initial mandatory mappers. M0, M1, M2, M3, M5 and M7. M4, M6, M8 and M9 are added after the core suite is stable; M9 may also serve as the second reproduced baseline.

Work items.

1. Build complete full-block output schema for every SiO_2 species-pair type.
2. Implement onsite global- $O(3)$ paths and offsite pair paths.
3. Implement strict parity-aware nonlinearities, norms, gates and residual scales.
4. Implement canonical Hermiticity and optional projection.
5. Implement full local- $O(2)$ bond-frame transforms and low- ℓ equivalence tests.
6. Build synthetic polynomial targets with known equivariant generating functions so every mapper can be tested independently of DFT noise.
7. Benchmark pair forward/backward time and memory over representative irrep sizes.

Acceptance gate.

Every mapper must satisfy: exact output dimension/type; no learned message-passing dependency; random global $O(3)$ equivariance; pair reversal/Hermiticity; stable float32 execution; and synthetic-target fitting. In float64 deterministic symmetry tests, relative equivariance error should be $\leq 10^{-8}$ for learned modules at random initialization and after training. For M7, arbitrary roll invariance and local/global low- ℓ path agreement are additional hard gates. Any architecture failing parity tests is excluded rather than evaluated only on proper rotations.

Expected outcome. M2 is the correctness/accuracy reference but expensive; M3 is simpler to audit; M7 should show the clearest high- L efficiency gain if the $O(2)$ bookkeeping is implemented correctly.

13.5 Stage 5 - Broad SiO_2 descriptor x mapper screening

Purpose. Identify the main accuracy/cost relationships on the published-protocol SiO_2 split.

Primary broad grid. Run the four descriptor families at their selected compact/high-information resolutions with the most informative mapper subset:

$$\{D1, D2, D3, D4\} \times \{M0, M1, M3, M5, M7\}. \quad (86)$$

Use one screening seed initially, then promote Pareto/non-dominated configurations to three-seed evaluation. Run dense M2, sparse M6, Gaunt M8 and kernel M9 on the best two or three descriptors, not indiscriminately on the full grid.

Mandatory controls.

- native ACE baseline from Stage 1;
- N-center linear/KRR as soon as M9 is complete;
- published HamGNN SiO₂ 2.29 meV as a clearly labelled contextual reference;
- matched parameter/throughput bands for M2/M3/M5/M7 comparisons;
- at least one matched descriptor-bytes comparison across D1-D4;
- learning curves for promoted configurations.

Acceptance gate.

Stage 5 passes when every reported model uses the identical frozen split, R_H , envelope, target mapper and test metric; all promoted learned models have at least three seeds; the complete result registry permits regeneration of every table; and no architecture choice is based on test data. The stage must identify at least one non-GNN accuracy baseline and an accuracy/storage/throughput Pareto set even if no new method beats published GNN numbers.

Expected outcome. D4 should improve low-data performance; M3 and M7 are expected to dominate the accuracy/cost frontier; dense M2 may achieve the lowest error among global-frame maps but at higher cost. A failure of D1/D2/D3 differences to affect Hamiltonian error would support the target-sufficiency-over-completeness interpretation rather than invalidate the descriptor analysis.

13.6 Stage 6 - Locality and target-sufficiency experiments

Purpose. Determine whether residual errors arise from mapper capacity, finite descriptor resolution, or true missing physics outside R_D .

13.6.1 R_D sweep

For the best descriptor/mapper families, repeat the controlled R_D sweep with descriptor resolution adjusted so angular/radial fidelity does not collapse at larger radius. Plot matrix MAE and target-sector errors against radius, descriptor bytes, and neighborhood size.

13.6.2 Outside-cutoff intervention test

For selected SiO₂ structures/pairs, hold every atom in

$$B(\mathbf{R}_i, R_D) \cup B(\mathbf{R}_j, R_D) \quad (87)$$

fixed, perturb only atoms outside the union, and recompute the DFT Hamiltonian using identical SCF/gauge settings. Since the pair descriptor input is exactly unchanged, any statistically significant ΔH_{ij} is an irreducible error for a strictly local model at that R_D .

Interventions should include remote neutral distortions, changes of distant layer/network geometry, and chemically legal remote substitutions only where total composition/state conventions remain meaningful. Repeat identical unperturbed calculations to estimate SCF/gauge noise.

13.6.3 Differential null-space test

For a smaller controlled subset, compute J_z and finite-difference/response J_h , remove rigid/global symmetry modes, and evaluate η_{miss} . Inspect the corresponding descriptor-null displacement modes and which Hamiltonian irreps respond.

13.6.4 Target collision analysis

Estimate a conditional target diameter

$$\Delta_H(\epsilon, \delta) = \sup \{ \|h(\mathcal{C}) - h(\mathcal{C}')\| : \|z(\mathcal{C}) - z(\mathcal{C}')\| \leq \epsilon, d_{\text{geom}}(\mathcal{C}, \mathcal{C}') \geq \delta \} \quad (88)$$

and the nearest-neighbor target spread in descriptor-pair space. Candidate descriptor collisions generated in Stage 3 should be evaluated with DFT where feasible to determine whether they are target-relevant.

Acceptance gate.

Stage 6 must produce a defensible locality conclusion, not merely the radius with lowest validation error. A recommended production R_D is accepted when outside-cutoff target changes and η_{miss} are below a significance threshold tied to the repeated-SCF/gauge noise and below a pre-registered fraction (target 10%) of the achieved model matrix-element MAE. If no practical radius reaches this level, the paper must conclude that the strictly local descriptor is insufficient for the desired accuracy and activate the long-range descriptor contingency or explicitly state the irreducible locality floor.

Expected outcome. SiO_2 should show a finite practical locality scale, but self-consistent electrostatic response may produce a nonzero long-range floor. The experiment is designed to reveal rather than suppress such a result.

13.7 Stage 7 - Controlled ablations and efficiency study

Purpose. Isolate which design choices create the final performance advantage.

Mandatory ablations.

1. full versus punctured versus full+punctured endpoint descriptors;
2. irrep-space versus matrix-space physical loss;
3. frozen range envelope versus no envelope (same target and optimizer budget);
4. natural-frequency sampling/weighting versus any introduced balancing scheme;
5. global dense $O(3)$ M2 versus factorized M5 versus complete local $O(2)$ M7 at matched hidden information and budget;
6. dense versus learned sparse paths M2/M6;
7. global CG versus Gaunt/Fourier M8 where ℓ_{max} is high enough to matter;
8. descriptor compact versus high-information resolution;
9. D1 versus D4 at fixed raw D1 content to quantify the value of deterministic covariant lifts;
10. exact canonical Hermiticity versus projection as a diagnostic;
11. optional generator order/body degree in M1/M3/M7.

Acceptance gate.

Each ablation changes one declared factor at a time or uses a factorial design whose interactions are explicitly analyzed. Runtime comparisons include precompute amortization: report descriptor build time, storage, training data throughput, pair inference throughput, and crossover point in number of epochs/inference evaluations. Any claimed speedup must be measured on identical hardware, precision, batch construction, and target schema.

Expected outcome. Range factorization should improve conditioning; $O(2)$ and factorized products should yield the largest mapper-side speed gains; D4 should trade storage/precompute for training sample efficiency; irrep/matrix losses should agree closely after metric alignment.

13.8 Stage 8 - Confirmatory robustness and external validation

Purpose. Test whether the conclusions survive cleaner structural separation, larger cells, and a second chemistry.

Work items.

1. Freeze a parent-polymorph-grouped SiO_2 split; rerun only the baseline and final Pareto models.
2. Evaluate the internal 150-atom bulk SiO_2 dataset zero-shot if target conventions match; otherwise re-train/evaluate as a separately labelled internal benchmark.
3. Port the native ACE/N-center baselines and best approximately 3 descriptor x 3 mapper combinations to monolayer MoS_2 .

4. Compare with published HamGNN/DeepH-E3/SLEM numbers using explicit split/dataset caveats.
5. Evaluate bands/eigenvalues/DOS for the final models under fixed electronic postprocessing.
6. Produce the final hypothesis ledger and Pareto-front conclusions.

Acceptance gate.

A principal conclusion about descriptor or mapper ranking is called robust only if it is not reversed without explanation on the grouped SiO₂ split and is qualitatively supported by either the internal large SiO₂ or MoS₂ confirmation. If a ranking is chemistry-specific, the final paper must state that explicitly. Published GNN numbers remain contextual unless the exact data/split/metric protocol is matched.

Expected outcome. Absolute errors should increase on the grouped split. The local pair architecture should remain size-transferable to the 150-atom cell if its local environments are covered. M7's speed advantage should become more pronounced on MoS₂ because of richer *d*-orbital/high-*L* targets.

14 Standard stage report template

Every stage closes with one versioned report. It should be concise enough to review quickly but exhaustive enough that another agent can reproduce the result without oral context.

1. **Stage objective and hypotheses.** List the exact hypotheses tested and the pre-registered expected outcomes.
2. **Frozen inputs.** Dataset versions, structure/split hashes, descriptor/envelope/target-schema hashes, code commit, hardware/software environment.
3. **Implemented work.** Deliverables completed, deviations from plan, and rationale for every deviation.
4. **Acceptance table.** Every gate item with numeric threshold, measured value, **PASS**/**FAIL**/**REVIEW**, and link/path to the raw verification artifact.
5. **Primary quantitative results.** Metrics with uncertainty and direct comparison to all relevant earlier baselines.
6. **Stratified results.** Irrep/species/distance/onsite-offsite breakdowns required by the stage.
7. **Symmetry and numerical verification.** Equivariance, inversion, Hermiticity, reconstruction, conditioning, gradient, or reference-agreement tests as applicable.
8. **Compute and storage.** Precomputation time, persistent bytes, training time, throughput, peak memory, parameter count and FLOP estimate.
9. **Issues and shortcomings.** Failed runs, numerical pathologies, discovered degeneracies, data problems, implementation compromises, and known untested cases.
10. **Hypothesis ledger.** For each hypothesis: supported, contradicted, unresolved, or not tested; include evidence, not only status.
11. **Decisions for the next stage.** Which configurations are promoted/dropped and why, stated before looking at later test sets.
12. **Reproduction manifest.** Exact command/config/registry IDs and artifact locations.

A stage may proceed with a non-critical **REVIEW** only if the report explains the risk and defines a follow-up. A critical **FAIL** in symmetry, target convention, reference agreement, leakage, or cache integrity blocks progression.

15 Failure modes and interpretation rules

15.1 Descriptor collision is not a mapper failure

If two pair inputs are identical within verified descriptor tolerance but their target blocks differ beyond DFT/gauge noise, no downstream deterministic mapper can solve the problem. Such a case must be reported as descriptor/target insufficiency, not addressed by increasing network depth.

15.2 Poor reconstruction does not by itself prove incompleteness

Inverse optimization can fail even for an injective descriptor. Formal conclusions require an analytic collision, rank deficiency, or theorem. Numerical reconstruction is evidence about stable practical invertibility, not a substitute for proof.

15.3 A complete descriptor can still be statistically hard to learn

Injectivity does not imply a well-conditioned inverse or a smooth low-complexity target map. Extremely small descriptor Jacobian singular values may make the target function steep in descriptor coordinates. This is why stable completeness and mapper capacity are measured separately.

15.4 Cutoff artifacts

A descriptor with a window that vanishes at R_D cannot claim injectivity for atoms at the boundary. Completeness claims use an interior margin. For learning, smooth cutoff windows are retained for continuity; boundary sensitivity is evaluated explicitly by moving atoms through the cutoff.

15.5 Gauge inconsistency

An electronic Hamiltonian matrix is representation-dependent. Before target-sufficiency conclusions, verify fixed AO ordering, real-harmonic convention, orbital phase, radial-shell ordering, periodic-image mapping, and any localization/orthogonalization convention. Rotate/permuted recomputations should match analytically transformed targets to the numerical DFT noise floor. If not, target gauge noise can masquerade as descriptor insufficiency.

15.6 Improper-rotation blind spots

Passing random $SO(3)$ tests is insufficient. Full Hamiltonian blocks contain pseudo irreps. Every descriptor and mapper is tested under inversion and improper reflections. Local-frame methods are not accepted until parity/reflection sectors are validated independently.

15.7 Leakage through preprocessing

Descriptor basis definitions may be fixed analytically, but any empirical scale, contraction, frequency selection, envelope fit, target normalization or feature pruning based on data must use training only. Test or validation information may not leak through precomputed global statistics.

15.8 Storage-driven cherry-picking

The 2 TB budget does not justify silently dropping difficult descriptor variants after seeing target performance. Candidate variants are pre-sized. If storage prevents a planned configuration, record the estimate and reason before Hamiltonian evaluation.

15.9 Published-number misuse

Literature values from GNN papers may use different splits, AO bases, target normalization, pair cutoffs or metrics. They are shown in a separate “published contextual” column unless exact comparability is demonstrated. The primary scientifically controlled comparisons are native methods sharing the same pipeline.

16 Required figures and tables for the final paper

The implementation project should produce the following publication artifacts automatically from the experiment registry.

1. Diagram of the no-message-passing pipeline: structure $\rightarrow$ cached atomic descriptors $\rightarrow$ pair input $\rightarrow$ full-block irrep output $\rightarrow$ envelope $\rightarrow$ AO block.
2. Table of complete target irreps and multiplicities for every dataset species-pair block type.

3. Descriptor table: irrep dimension, bytes/atom, precompute rate, reconstruction success, Jacobian singular-value quantiles, theorem status.
4. Descriptor basis-convergence plots versus dimension and bytes.
5. Collision/reconstruction examples, including at least one known incomplete-invariant negative control.
6. Target-sufficiency plots: outside-cutoff response versus R_D , η_{miss} , and nearest-neighbor target spread.
7. Full descriptor x mapper SiO_2 matrix-element MAE heatmap with missing expensive cells clearly marked.
8. Accuracy versus pair throughput Pareto front and accuracy versus total descriptor storage Pareto front.
9. Learning curves for ACE, N-center, and promoted D1-D4/M3/M7 models.
10. Error versus distance and envelope magnitude.
11. Error by target L, p , with pseudo irreps explicitly highlighted.
12. Global- $\text{O}(3)$ versus local- $\text{O}(2)$ accuracy/cost table at matched budgets.
13. Random versus parent-grouped SiO_2 split comparison.
14. 150-atom SiO_2 and MoS_2 confirmatory tables.
15. Final hypothesis ledger with evidence references.

17 Decision criteria for the final recommended method

No single metric determines the winner. A final production recommendation must be Pareto-nondominated and satisfy all hard mathematical/symmetry gates. Rank candidate models using the following ordered logic:

1. eliminate any model with incorrect $\text{O}(3)$ equivariance, pair reversal, gauge inconsistency, or target leakage;
2. eliminate any descriptor whose demonstrated target-insufficiency exceeds the desired error scale at the chosen R_D ;
3. among remaining methods, identify the minimum matrix-element MAE and its uncertainty;
4. retain all methods statistically indistinguishable in accuracy within a pre-registered tolerance (suggested 5% relative MAE or paired-bootstrap overlap);
5. among that set, prefer lower end-to-end cost including descriptor precomputation/storage and pair inference;
6. break close ties in favor of simpler mathematical auditability and better grouped/OOD robustness.

The paper should therefore be able to conclude, for example, that one descriptor is best for theorem strength, another for storage, and another for absolute predictive accuracy, while still selecting one recommended production combination.

A Notation and irrep bookkeeping

Symbol	Meaning
$\mathcal{C}$	periodic atomic configuration plus fixed electronic-state protocol
$\mathcal{N}_i$	atom i neighborhood inside descriptor cutoff R_D
R_H	Hamiltonian pair cutoff established from target data
R_D	descriptor neighborhood cutoff, independently swept
$\mu_i^{(s)}$	species- s finite point measure around atom i
D_i	precomputed typed atomic descriptor
$\mathbf{r}_{ij}^{\mathbf{L}}$	directed pair displacement including lattice translation
$H_{ij}^{\mathbf{L}}$	complete AO Hamiltonian block
$\mathbf{h}_{ij}^{\mathbf{L}}$	complete MANDALA block-irrep vector
(ℓ, p)	global $\text{O}(3)$ angular momentum and inversion parity
q	full-block irrep multiplicity index; may encode shell/radial provenance internally

Symbol	Meaning
m	magnetic component within one (ℓ, p) irrep copy
Q_{ij}	fixed AO-block to irrep transformation
S_{ij}	fixed irrep-space pair-reversal/Hermiticity transform
$G_{ij}(r)$	frozen range-magnitude scaling transform
z_{ij}	complete pair input used in target-sufficiency tests
J_D, J_z, J_h	descriptor, pair-input, and Hamiltonian Jacobians

B Full-block target construction algorithm

The coding agent should implement the target mapper from AO metadata, not handwritten target lists.

Algorithm A1: build full-block irrep schema.

1. Read the canonical ordered AO basis for species A and B : each basis function has radial-shell identifier, angular momentum ℓ , and real-harmonic component m .
2. Partition AO rows/columns into radial-shell angular blocks (a, b) while preserving the original AO matrix ordering.
3. For each (a, b) construct the fixed real-basis CG/Wigner transform from the Cartesian AO subblock to

$$\bigoplus_{L=|\ell_a-\ell_b|}^{\ell_a+\ell_b} L^{(-1)^{\ell_a+\ell_b}}.$$

4. Concatenate all subblock irrep outputs into one species-pair full-block schema in a deterministic order. Equal (L, p) occurrences remain separate multiplicity copies.
5. Generate an inverse transform back to the exact original AO row/column order.
6. Generate the reverse-pair schema mapping by applying AO transpose followed by the forward irrep transform; simplify to an index/sign/orthogonal block mapping and cache it.
7. Unit-test random round trips, random $O(3)$ actions, transpose/pair reversal, and norm preservation.

The scientific model never predicts the subblock outputs separately. Step 2 exists only to construct the mathematically correct direct-sum basis.

C Descriptor algorithms

C.1 D1 pseudocode

```

for atom i:
  coeff = zeros(species, radial_n, l, 2*l+1)
  for neighbor a in N_i(R_D):
    s = species(a)
    rvec = displacement(i, a, image)
    r = norm(rvec)
    for l <= l_max:
      Y = real_spherical_harmonics(l, rvec/r)
      for n <= n_max(l):
        coeff[s,n,l] += w(r)*R[n,l](r)*Y
  store coeff according to declared irreps

```

For $r = 0$ contributions only $\ell = 0$ are valid; ordinary atomic neighborhoods exclude the center itself unless an explicit center/species channel is defined.

C.2 D2 pseudocode

```

for atom i:

```

```

    for neighbor a:
        rtilde = displacement / R_D
        evaluate orthogonal radial/Zernike recurrence
        evaluate real solid harmonics / STF components
        accumulate species-resolved moment irreps
return ordered direct sum of all selected (n,l,p) copies

```

The implementation should expose both an irrep view and, on test systems, a Cartesian moment view whose fixed linear conversion is verified numerically.

C.3 D3 pseudocode

```

for atom i:
    for neighbor a:
        r, rhat = geometry(a)
        Y_lm = spherical_harmonics(rhat)
        for frequency q_k:
            for l <= l_max:
                F[s,k,l] += w(r)*spherical_bessel_j(l,q_k*r)*Y_lm[l]
return real parity-labelled irrep array

```

Frequency grids must be versioned. Uniform- q , quadrature-inspired, and theorem-derived sample sets are separate configurations.

C.4 D4 pseudocode

```

raw = D1
lifted = []
for deterministic coupling_path in path_manifest:
    y = clebsch_gordan_contract(raw[path.a], raw[path.b], path.L)
    if path.degree == 3:
        y = clebsch_gordan_contract(y, raw[path.c], path.out_L)
    lifted.append(y)
return concatenate(raw, lifted)

```

The path manifest is generated solely from declared basis/angular rules and stored with a hash.

D Pair-mapper algorithms

D.1 Reduced-coefficient M3

Algorithm A2.

1. Convert D_i, D_j to the configured full/punctured pair input.
2. Evaluate fixed bond radial/spherical basis b_{ij} .
3. Evaluate a versioned generator manifest of CG coupling trees producing only target full-block irrep types.
4. Evaluate true 0e invariant contractions and invariant norms. No raw pseudoscalar is passed to an unrestricted MLP.
5. Concatenate invariant chemistry/range metadata and predict all scalar reduced coefficients with one or more ordinary MLPs.
6. Multiply and sum generator copies into every target multiplicity q .
7. Apply canonical pair reversal construction as configured.
8. Apply $G_{ij}(r)$ and return the physical full-block irrep vector.

D.2 Bond-frame M7

Algorithm A3.

1. Construct a deterministic bond frame from $\hat{r}_{ij}$; store no arbitrary global orientation in the learned input.
2. Rotate global endpoint irreps with exact real Wigner-D matrices.
3. Convert/reindex to local O(2) representations while retaining parent (ℓ, p, q) metadata.
4. Apply O2Linear/O2TensorProduct/O2Gate operations. Reflection parity is explicit.
5. Produce the complete target local representation, convert to the requested global target irreps and rotate back.
6. Verify output independence under an injected random roll ϕ by applying the corresponding O(2) gauge transform internally.
7. Enforce pair reversal and range envelope.

E Symmetry and numerical test catalogue

All tests below are automated and run in CI on the small Si dataset and synthetic random irreps.

Test	Definition	Initial target
Descriptor SO(3)	random rotations, $D(QN)$ vs $\rho(Q)D(N)$	10^{-9} f64
Descriptor inversion	$D(-N)$ vs declared parity action	10^{-9} f64
Improper reflection	random $\det = -1$ orthogonal matrices	10^{-9} f64
Permutation	reorder equal-species neighbors	exact/f64 tol.
PBC image	equivalent cell/image representation	exact/f64 tol.
Puncture identity	recompute-without-neighbor vs subtraction	10^{-10} f64
AO-irrep roundtrip	random block forward/inverse transform	10^{-10} rel.
Pair reversal	irrep S_{ij} vs AO transpose	10^{-10} rel.
Mapper O(3)	$F(Qz)$ vs $\rho_H(Q)F(z)$	10^{-8} f64
Mapper post-train	same after optimization	10^{-8} f64
Bond roll	inject random local $R_z(\phi)$ gauge	10^{-8} f64
O2 vs CG microtest	exhaustive low- ℓ selected products	10^{-9} f64
Hermiticity	reconstructed $H_{ji} - H_{ij}^T$	10^{-10} f64 ex-act map
Loss isometry	irrep metric vs AO Frobenius	10^{-10} rel.
Cache repeatability	regenerate same shard/metadata	bitwise or documented tol.
No-message-passing	output insensitive to irrelevant graph metadata	bitwise deterministic

Initial numerical targets may be relaxed only for a documented conditioning reason and must remain far below the meV scientific error scale.

F Initial hyperparameter and resolution search spaces

These are starting search spaces, not immutable scientific choices. Changes must be registered before test evaluation.

F.1 Descriptors

- R_D : four or more data-informed radii anchored at R_H as specified in the main text.

- D1: $\ell_{\max} \in \{3, 4, 5, 6\}$; radial channels chosen so total bytes span compact to high-information regimes; compare two orthogonal radial bases.
- D2: maximum polynomial/Zernike total degree selected from a convergence sequence, e.g. 4, 6, 8, 10, with automatic removal of algebraically redundant moments.
- D3: $\ell_{\max} \in \{3, 4, 5, 6\}$ and multiple radial frequency counts; include at least one frequency set derived from a finite-recovery argument if available.
- D4: retain D1; body degree 2 initially, then selected degree 3; cap path count by deterministic angular/radial budgets and storage estimator.

F.2 Mappers

- Scalar MLP widths: approximately 64, 128, 256 with 2–5 hidden layers, then adjusted to matched parameter bands.
- Global hidden $\ell_{\max}$: target-reachability minimum through 6; no higher value without an ablation justification.
- M3 generator degree: pair/bond degree 1–3 initially; intermediate K paths exhaustive within selected cap before pruning.
- M5 factorization rank: logarithmic/small set from highly factorized to near-dense reference.
- M6 path target sparsity: 25%, 50%, 70%, 85% retained/pruned regimes with dense warm-up.
- M7 local $O(2)$ depth: 1–4 blocks; invariant MLP matched to M3 parameter budgets.
- M8 bandlimit matched to global target L ; parity-doubled implementation required.

F.3 Optimization

A common optimizer/scheduler family should be used across neural mappers unless a method has a documented stability requirement. Initial defaults: AdamW, mixed precision only after float32 parity tests pass, gradient clipping logged but not enabled unless necessary, early stopping on validation physical MAE with a fixed maximum epoch budget. Optimizer hyperparameters are tuned on a small representative subset and then frozen by mapper family to avoid giving individual descriptor combinations unequal tuning effort.

G Storage budget calculation

For descriptor d with K_d float components per atom, dtype size b bytes, average atoms/structure $\bar{N}$, and S structures,

$$B_d \approx S \bar{N} K_d b / c_d, \quad (89)$$

where $c_d \geq 1$ is any measured lossless compression factor. Do not assume a compression factor before measuring it. For multiple R_D and resolutions, sum actual distinct cache sizes; when one larger-radius additive cache can losslessly derive smaller radii, store the derivation only if it does not alter the descriptor window definition.

Before materializing a cache, the pipeline produces a budget table:

Artifact	count	dtype	raw GB	est. GB	retention policy
neighbor oracle					keep if feasible
D1 caches					compact + high-res
D2 caches					compact + high-res
D3 caches					compact + high-res
D4 caches					selected path sets
target irreps					mandatory
analysis/Jacobians					sample only
checkpoints/logs					top-k + final

The automated hard-stop threshold is 1.8 TB unless a project review reallocates the 10% reserve.

H Theorem and proof obligations

The paper must distinguish theorem, proposition under cited assumptions, and empirical evidence. The following proof obligations are required.

H.1 D1 complete-basis injectivity

State the function space and radial/angular completeness assumptions; prove reconstruction of each species density from all coefficients; then show recovery of the finite species-labelled measure for delta or injective smoothing. State the cutoff positivity/interior condition explicitly.

H.2 D2 finite-domain result

Specify $N_{\max}$, support ball, minimum separation/genericity assumptions, moment index set, and species treatment. Reduce the collected irreducible moments by an explicit invertible linear transform to a moment set covered by a multivariate finite-measure/Prony theorem. State whether uniqueness is global, generic, or local and whether the theorem gives stability bounds.

H.3 D3 finite-sampling result, if claimed

Provide an explicit map between selected spherical Fourier-Bessel coefficients and a sufficient finite Cartesian Fourier sample set. A statement that the continuous Fourier transform is injective is insufficient to prove a finite-shell descriptor injective.

H.4 D4 inheritance

If D4 contains D1 unchanged, injectivity follows immediately because equality of D4 implies equality of its D1 subvector. This simple proof should be stated rather than implying the nonlinear lifts independently establish completeness.

H.5 M3 equivariance

Prove that invariant coefficients multiplying target-irrep generators transform equivariantly, as in Proposition 2. Verify all computed “invariants” are truly 0e under full $O(3)$, not merely rotation-invariant pseudoscalars.

H.6 M7 global equivariance

State the local $O(2)$ representation, frame transition function, and stabilizer-equivariance condition. Prove independence of frame roll and global $O(3)$ covariance. Treat frame-chart singularities numerically with a continuous/robust convention and show they do not create output discontinuities.

I Target-sufficiency experiment details

I.1 DFT intervention noise floor

For every structure chosen for intervention, first repeat the unmodified SCF calculation at least three times or under controlled restart variations to quantify numerical/gauge variability. Define a block-level noise distribution σ_{SCF} . A remote intervention is considered target-relevant only when its transformed full-block change exceeds both a multiple of this noise floor and a practical meV threshold.

I.2 Remote perturbation construction

An intervention at descriptor radius R_D must leave the exact cached pair input unchanged to numerical precision. After modifying remote atoms:

1. rebuild the descriptor/neighborhood and verify $\|\Delta z\| < \epsilon_z$;
2. rerun the fixed electronic-structure protocol;
3. align/check AO gauge;

4. evaluate $\|\Delta h\|$ globally and by target irrep;
5. compare to repeated-SCF controls.

This experiment directly estimates the irreducible error floor of any strictly local pair mapper at that radius.

I.3 Jacobian computation

For J_h , use central finite differences where DFPT is unavailable:

$$\frac{\partial h}{\partial q_k} \approx \frac{h(q_k + \Delta) - h(q_k - \Delta)}{2\Delta}. \quad (90)$$

Perform a displacement-step convergence study on a small subset before generating the full Jacobian. Work in a coordinate basis with rigid translations removed; global rotations may be handled by covariant comparison or projected out when testing descriptor-null internal distortions.

I.4 Nearest-neighbor sufficiency diagnostic

Normalize descriptor-pair distance using train-only per-irrep scales and an $O(3)$ -aware common-frame comparison. Search close pair inputs in the dataset, transport targets into a common frame if necessary, and plot target difference versus descriptor distance. A non-vanishing conditional target spread as descriptor distance approaches the numeric resolution indicates either target insufficiency, gauge inconsistency, or descriptor ill-conditioning; investigate in that order.

J Baseline reproduction details

J.1 ACEhamiltonians

Reference the upstream ACEhamiltonians.jl implementation only for controlled feature/prediction agreement. The native implementation should reproduce the mathematical basis, symmetrization and regression while conforming to MANDALA tensor conventions. Where upstream uses a different spherical-harmonic normalization/order, build an explicit fixed conversion and verify physical predictions rather than comparing raw arrays before alignment.

Use released AI examples to verify the published qualitative behavior. The final project baseline, however, is the native model trained on HamGNN SiO₂ with the common split, R_H , envelope and full-block metrics. Original AI error numbers are never placed in the same numeric performance column as SiO₂.

J.2 N-center linear/KRR

Implement pair-tagged density-correlation features with explicit atom-index exchange sectors and target parity. First reproduce an upstream/public molecular Hamiltonian example where possible, then train on SiO₂. Linear regression provides the cleanest representation baseline; KRR tests whether a nonlinear invariant similarity function closes part of the gap without introducing equivariant neural hidden states.

K Bibliography and primary references

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L Final implementation checklist

Before declaring the project implementation complete, verify every item below.

- ☐ Full-block prediction interface is used for every model; no shell-specific scientific heads exist.
- ☐ Spinless/non-SOC target scope is enforced in datasets and representation code.
- ☐ Canonical MANDALA/OpenMX target gauge is frozen and tested.
- ☐ Native ACE baseline is reference-validated and evaluated on SiO₂ before new-model claims.
- ☐ D1-D4 and ragged oracle caches have exact irrep metadata and content hashes.
- ☐ At least one finite descriptor has a domain-qualified theorem attempt/result; all descriptors have explicit completeness status.
- ☐ Reconstruction, collisions, Jacobian rank and stable-conditioning analyses are complete.
- ☐ M0-M9 implemented or explicitly documented as dropped with reason; mandatory core models M0, M1, M2, M3, M5 and M7 completed.
- ☐ Local O(2) implementation passes parity, reflection and roll-invariance tests.
- ☐ R_H established from training targets; R_D swept independently.
- ☐ Target-sufficiency/outside-cutoff tests establish a locality conclusion.
- ☐ Frozen range envelope shared by all methods within a dataset.
- ☐ Primary training is in irrep space; physical matrix-element MAE always reported.
- ☐ Irrep-vs-matrix loss ablation completed later without changing the main study focus.
- ☐ Total persistent storage remains within 2 TB with reserve policy documented.
- ☐ Random HamGNN-style SiO₂ split used first; grouped split performed only in final robustness stage.
- ☐ Internal 150-atom SiO₂ and public MoS₂ confirmation completed for final Pareto models.
- ☐ GNN results are clearly marked published/contextual and not presented as reproduced.
- ☐ Every stage has a finalized report with acceptance table, baseline comparison, issues and shortcomings.
- ☐ Final hypothesis ledger distinguishes supported, contradicted and unresolved claims.